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H^1 stability and convergence analysis of the L1/L1-2 Legendre spectral method for non-local weakly singular integro-PDEs

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Abstract
This work deals with a numerical approach based on a nodal Galerkin spectral method, which is a proper combination of the Legendre spectral method and the L1/L1-2 discretizations to address a class of weakly singular integro-partial differential equations in one and two dimensions having non-local temporal operators. Such equations are fundamental in modeling processes that exhibit memory and hereditary properties, often encountered in fields like anomalous diffusion and viscoelasticity. The primary goal of this work is to develop a stable high-order numerical scheme that can efficiently solve these problems. The numerical approach follows by combining the temporal semi-discretization and the spectral method that involves both trial and test functions being chosen as the Lagrange interpolating polynomials defined at the Legendre–Gauss–Lobatto points. In addition, we investigate their effect on numerical estimates. The L1 or L1-2 scheme is used to discretize the time-fractional operator of the semi-discrete problem, whereas to deal with the weakly singular Volterra operator, a modified quadrature rule (which is a combination of the composite trapezoidal approximation and the midpoint rule) is employed. In contrast, the Legendre spectral method is utilized for an efficient approximation in the space directions for 1D and 2D problems, for which the rate of convergence in space increases with respect to the smoothness of the solution. This spectral approximation involves the order of Legendre polynomials. In addition, it is numerically observed that the fully discretized approximation leads to a stable and convergent numerical solution for problems having an initial layer in time, which remains reliable regardless of the chosen time step size. Moreover, this hybrid numerical approach leads to higher-order accuracy in space and time, where the accuracy rate depends on the degree of the polynomial approximation, the regularity of the solution space, and the order of the weakly singular kernel. Theoretical findings are supported by extensive numerical experiments of various complex fractional integro-differential equations having weakly singular kernels

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with different regularity conditions on the exact solutions, and it shows the strong effectiveness of the present approach.

Keywords Legendre-Spectral method · 1D and 2D Integro PDEs · Weakly-singular kernels · Nonlocal problems in time · Galerkin spectral method · Convergence analysis · Weak formulation · Integral equations · Stability estimate · AMS MathSciNet Classifications: 35K20, 35R09, 45K05

1 Introduction

Time-fractional parabolic integro-partial differential equations (TFIPDEs) with singular kernels are increasingly significant in various real-life applications due to their ability to effectively model complex systems characterized by memory effects and non-local interactions [1]. Specifically, memory effects refer to the system's dependence on its entire history, modeled via the time-fractional derivative, which integrates past states with a power-law kernel. Non-local interactions can also arise from the integral terms in the model, capturing spatial or spatio-temporal influences over a region rather than at a single point. These features are essential in accurately describing complex physical and biological processes and are incorporated through fractional and integral operators. In materials science, these equations are used to describe heat transfer in heterogeneous materials, such as composites or porous media, where thermal conductivity may depend on the history of temperature distributions, reflecting intricate internal structures that influence heat flow [2, 3]. This is particularly relevant in engineering applications where accurate modeling of heat dissipation is critical to the performance and safety of materials under thermal stress. In finance, time-fractional derivatives play a crucial role in modeling asset price dynamics, capturing the phenomenon of volatility clustering and heavy tails observed in financial markets. For instance, see the work given in [4]. These models can incorporate the impact of past price movements on current values, providing a more nuanced understanding of market behavior, which is essential for risk management and derivative pricing. Furthermore, in biological systems, TFIPDEs are used to model the spread of infectious diseases, where the dynamics of disease transmission are influenced by the history of interactions within a population, allowing for the design of more effective intervention strategies; the singular kernels in these models can represent the influence of past contacts on current infection probabilities, making them invaluable in public health planning [5, 6]. The study of viscoelastic materials [7] indeed benefits substantially from fractional calculus, especially through time-fractional partial differential equations (TFPDEs), which help to characterize complex stress–strain relationships that exhibit time-dependent phenomena like creep and relaxation. These behaviors are central in materials science, where the long-term stability and performance of materials under stress are critical applications that range from biomedical devices to structural components.

Weakly singular integral parts can be considered as a generalized extension of Abel integrals with non-local behavior throughout the time interval. It also holds significant importance across various domains of scientific computing, as it is the

initial step before considering most fractional derivatives. For these weakly singular problems, it is observed that the well-posedness of the continuous problem is not straightforward, as the solution may not exist in its domain of definition. In particular, the existence of the solution also depends on the fractional order derivative [8, 9]. This restriction makes it complicated to produce an approximate solution and its convergence analysis. In this context, Volterra-Fredholm integral operators and also nonlinear problems are the most interesting ones, which are considered in [10, 11]. For example, Luo et al. [12] developed numerical methods for time-fractional integro-differential equations, effectively capturing the dynamics of materials with intrinsic time-dependent responses, which is essential for modeling processes in viscoelastic and other complex materials. Similarly, [13] proposed compact finite difference schemes tailored for weakly singular integro-differential equations, highlighting their capacity to model physical systems where the singular behavior influences the stability and precision of solutions. Furthermore, work by Luo et al. [14] on compact difference schemes addresses equations with weakly singular kernels, reinforcing the significance of these equations in applications like heat transfer within heterogeneous media and diffusion in porous structures. The versatility of time-fractional parabolic integro-differential equations, particularly with singular kernels, demonstrates their role in enriching our understanding of real-world phenomena and bridging the gap between theory and practice across disciplines. As research progresses, the capacity of these equations to enhance our grasp of dynamical systems and refined predictive modeling offers promising avenues for new applications and advancements in science and engineering. Fractional integro-differential equations with weakly singular kernels; further applications are found in fields such as heat conduction, radiative equilibrium, elasticity, and fracture mechanics [15]. Currently, fractional calculus is undergoing rapid developments with more applications in the fields, such as image processing, computer vision, and the dynamics of the bank data, which can be seen in [16, 17].

Over the past decades, several researchers have worked on numerical solutions based on polynomial or collocation-based fractional integro-differential equations (FIDEs). For example, [18] used the Legendre wavelet technique for the numerical solution of FIDEs. In this technique, the author converted the FIDEs into a system of linear equations. Using the derived system, the author found an approximate solution of FIDEs. [19] and [11, 20] found approximate solutions of the FIDEs using the Taylor expansion with the Legendre wavelet method and homotopy methods, respectively. Legendre wavelets were also used in [21] to solve weakly singular Fredholm integro-differential equations. [22] found the numerical solution of FIDEs by using the fractional differential transform method. [23] used the operational Tau approximation method for the solution of FIDEs. Haar and Hermite wavelets are also used for FIDE in [24–26].

A crucial aspect of solving fractional integro-differential equations, especially those involving weakly singular kernels, is characterized by their singularity at a certain point, usually at the lower limit of the integral. Unlike regular kernels, these kernels present computational difficulties due to their non-smooth behavior. However, the inclusion of weakly singular kernels is essential for accurately modeling real-world phenomena, particularly in systems with fading memory or hereditary

properties. For example, in fractional differential models of epidemic phenomena in biology, fluid mechanics, or gas dynamics, the singularity of the kernel reflects the gradual loss of influence of the system from past states. Thus, the development of efficient numerical methods that handle such weak singularities is of great importance in these applications [27–29]. In addition efficient ways of layer behaviors in space are also tackled for several problems including coupled systems, like in [30–33]. We observe that the numerical procedure for problems having weakly singular kernels is less accurate than that for problems with regular kernels. In addition, we must note that the convergence analysis of numerical approaches for problems having unbounded solution derivatives are also addressed in several works, like [34, 35] on space-time adaptive meshes. One can also avoid the use of derivatives a priori in the analysis by using a posteriori approach given in [36]. Furthermore, a widespread application of the weakly singular kernels, including fractional-order sub-diffusion models and their various approximation techniques based on the Legendre spectral method, can be found in [37–40] and references therein.

Partial integro-differential equations (PIDEs) are a generalized form where all of these operators involve integral and fractional operators, and the classical derivatives are present, and their appearances can also be observed in the applied models [15]. More specifically, some phenomena in chemistry, hydrology, fluid mechanics, physics, gas dynamics, and signal processing are best described by using partial differential equations [41] and time-fractional partial integro-differential equations whose numerical analysis is available in few cases, such as in [42] through the reproducing kernel algorithms. Most of the existing methods for solving fractional partial integro-differential equations (FPIDE) consist of an integral operator with respect to time, where the kernels are mainly considered as convolution kernels; see [43, 44]. The present objective is to consider a kernel that is of weakly singular type.

It is still a challenging task to develop a simple and efficient numerical method for time-fractional integro-partial differential equations (TFIPDEs) involving weakly singular kernels. In this paper, we propose an efficient method that combines the $L1$ or $L1$-2 schemes in time with the Legendre spectral method in space for solving such weakly singular TFIPDEs, which can be expressed in the form:

$$\begin{cases} \dfrac{\partial^{\alpha} u\,(\boldsymbol{x},t)}{\partial t^{\alpha}} - \mathcal{L}u(\boldsymbol{x},t) = f(\boldsymbol{x},t) - \lambda \displaystyle\int_0^t \mathbb{K}(\boldsymbol{x},t,s)u(\boldsymbol{x},s)ds, \\ \text{where } (\boldsymbol{x},t) \in \Omega := \Omega_{\boldsymbol{x}} \times \Omega_t, \quad \text{with} \\ \text{initial conditions : } u(\boldsymbol{x},0) = \phi(\boldsymbol{x}), \quad \text{for all} \quad \boldsymbol{x} \in \overline{\Omega}_{\boldsymbol{x}}, \\ \text{and boundary conditions : } u(\boldsymbol{x},t) = \psi(\boldsymbol{x},t), \quad \text{for all} \quad (\boldsymbol{x},t) \in \partial\Omega_{\boldsymbol{x}} \times \overline{\Omega}_t, \end{cases} \tag{1}$$

where $\Omega_t = (0,T]$, $\Omega_{\boldsymbol{x}} \subsetneq \mathbb{R}^d$ $(d = 1, 2)$ and $\partial\Omega_{\boldsymbol{x}}$ denote its boundary. We consider f to be a sufficiently smooth function on $\overline{\Omega} = \Omega \cup \partial\Omega$ and $\frac{\partial^{\alpha} u(\boldsymbol{x},t)}{\partial t^{\alpha}}$ to denote the Caputo-type time fractional differential operator of order α, defined later. Here $\mathcal{L}u(\boldsymbol{x},t) = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}$ for $d = 2$ and $\mathcal{L}u(\boldsymbol{x},t) = \frac{\partial^2 u}{\partial x^2}$ for $d = 1$. In addition, we consider the input data f, ϕ, ψ to satisfy the required compatibility conditions; in order to have sufficient regularity on the solution for the present convergence analysis. The weakly singular kernel $\mathbb{K}(\boldsymbol{x},t,s)$ is considered to have the form:

$$\mathbb{K}(\boldsymbol{x},t,s)=K(\boldsymbol{x},t)(t-s)^{-\beta},\ \beta\in(0,1),$$

where $K(\boldsymbol{x},t)>0$ is a sufficiently smooth continuous function, defined in $\overline{\Omega}$. While the present research concentrates on time-fractional weakly singular integro-partial differential models with a single spatial variable (i.e., $d=1$), our findings reveal that the proposed L1/L1-2 Legendre spectral method performs exceptionally well when extended to two-dimensional problems.

The paper is structured as follows. Section 2 describes the development of semi-discrete temporal schemes utilizing the L1 and L1-2 techniques for the time fractional derivative as well as for the integral operator of Volterra type. Section 3 provides the stability and convergence analysis for the time semi-discretization of the present problem. Section 4 introduces the Legendre spectral technique. Section 5 investigates the impact of the combination of space and time discretizations along with their convergence properties. The numerical experiments presented in Sect. 6 offer practical examples to validate the theoretical findings. Finally, Sect. 7 presents the general conclusions.

Notation: To introduce the variational formulation that will be required for spectral approximation, we introduce some functional spaces endowed with standard norms and inner products, $H^1(\Omega_x):=\left\{v\in L^2(\Omega_x),\frac{\partial v}{\partial x}\in L^2(\Omega_x)\right\}$,$H_0^1(\Omega_x):=\left\{v\in H^1(\Omega_x),\,v|_{\partial\Omega_x}=0\right\}$, $H^m(\Omega_x):=\left\{v\in L^2(\Omega_x),\frac{\partial^k v}{\partial x^k}\in L^2(\Omega_x),\quad \text{for all positive integer } k\leqslant m\right\}$ where $L^2(\Omega_x)$ is the space of measurable functions whose square is Lebesgue integrable in Ω_x. The inner products of $L^2(\Omega_x)$ and $H^1(\Omega_x)$ are defined, respectively, by $(u,v)=\int_{\Omega_x}uv\,\mathrm{d}x,\quad (u,v)_1=(u,v)+\left(\frac{\partial u}{\partial x},\frac{\partial v}{\partial x}\right)$, and the corresponding norms by $\|v\|=(v,v)^{1/2},\quad \|v\|_1=(v,v)_1^{1/2}$. The norm $\|\cdot\|_m$ of the space $H^m(\Omega_x)$ is defined by $\|v\|_m=\left(\sum_{k=0}^m\left\|\frac{\partial^k v}{\partial x^k}\right\|^2\right)^{1/2}$. In this article, instead of using the above standard H^1 norm, we prefer to define $\|\cdot\|_1$ by $\|\cdot\|_1$

$$\|v\|_1=\left(\|v\|^2+\alpha_0\left\|\frac{\partial u}{\partial x}\right\|^2\right)^{1/2},\tag{2}$$

and use it throughout the present analysis, where the specific expression of α_0 will be defined later.

2 Temporal semi-discretization

The time fractional parabolic integro-differential equations (1) and the accompanying time semi-discretizations are studied in this section with truncation error bounds. To simplify the notation and for the streamlined construction of the scheme in a temporal direction, we consider the special case $f=0$, although the fully discretized problem through the Legendre spectral approach involves a general source function. Needless

to mention that the present numerical approach works for general problems, as shown in the numerical section. Consider $N \in \mathbb{N}$. Now let us define the uniform mesh in time, which is given by

$$\Omega_t^N := \{t_k \mid t_k = k\Delta t, \Delta t = T/N, \text{ for } k = 0, 1, \ldots, N\}. \tag{3}$$

2.1 Semi-discretization using L1 technique

For $k = 0, 1, ..., N-1$, the Caputo fractional derivative $\frac{\partial^\alpha u}{\partial t^\alpha}$ at $(\cdot, t_{k+1})$, where $t_{k+1} \in \Omega_t^N$, can be expressed as:

$$\frac{\partial^\alpha u\,(x, t_{k+1})}{\partial t^\alpha} = \frac{1}{\Gamma(1-\alpha)} \sum_{j=0}^{k} \int_{s=t_j}^{t_{j+1}} (t_{k+1} - s)^{-\alpha} \frac{\partial u}{\partial s}(x, s) ds.$$

This can be approximated by the traditional L1 scheme [45] as follows.

$$\begin{aligned} \frac{\partial^\alpha u\,(x, t_{k+1})}{\partial t^\alpha} =& \frac{1}{\Gamma(1-\alpha)} \sum_{j=0}^{k} \frac{u(x, t_{k+1-j}) - u(x, t_{k-j})}{\Delta t} \int_{s=t_j}^{t_{j+1}} (t_{k+1} - s)^{-\alpha} ds + \Phi_{L1}^{k+1} \\ =& \frac{(\Delta t)^{-\alpha}}{\Gamma(2-\alpha)} \sum_{j=0}^{k} \left[u(x, t_{k+1-j}) - u(x, t_{k-j})\right] \left((j+1)^{1-\alpha} - j^{1-\alpha}\right) + \Phi_{L1}^{k+1} \\ \equiv& \mathcal{D}_{L1}^\alpha u(x, t_{k+1}) + \Phi_{L1}^{k+1}, \end{aligned} \tag{4}$$

where Φ_{L1}^{k+1} is the truncation error bound of the time fractional operator based on $L1$ scheme. For the sake of simplicity, let us introduce the notation.

$$b_j = (j+1)^{1-\alpha} - j^{1-\alpha} \text{ for } j = 0, 1, ..., N-1. \tag{5}$$

Then, we have $\mathcal{D}_{L1}^\alpha u(x, t_{k+1}) = \frac{(\Delta t)^{-\alpha}}{\Gamma(2-\alpha)} \sum_{j=0}^{k} \left(u(x, t_{k+1-j}) - u(x, t_{k-j})\right) b_j$. The integral term in the model problem at the point $(\cdot, t_{k+1})$ for $k = 0, 1, ..., N-1$ can be expressed as

$$\begin{aligned} \mathcal{J}_x^t u(x, t_{k+1}) \equiv& \int_{t_0}^{t_{k+1}} \mathbb{K}(x, t_{k+1}, s) u(x, s) ds \\ =& \sum_{j=0}^{k-1} \int_{t_j}^{t_{j+1}} \mathbb{K}(x, t_{k+1}, s) u(x, s) ds + \int_{t_k}^{t_{k+1}} \mathbb{K}(x, t_{k+1}, s) u(x, s) ds. \end{aligned}$$

To discretize the integral operator, it is important to note that the kernel $\mathbb{K}$ has a singularity at $s = t_{k+1}$. In order to tackle this issue, the composite trapezoidal approximation is employed on the interval $[t_0, t_k]$, and on the final interval $[t_k, t_{k+1}]$, we use a modified version of the midpoint rule, which is described as follows [46]:

$$\begin{aligned}\mathcal{J}_x^t u(x,t_{k+1}) &= \frac{\Delta t}{2}\sum_{j=0}^{k-1}\left[\mathbb{K}(x,t_{k+1},t_j)u(x,t_j)+\mathbb{K}(x,t_{k+1},t_{j+1})u(x,t_{j+1})\right]\\ &+\frac{(\Delta t)^{1-\beta}}{2(1-\beta)}K(x,t_{k+1})(u(x,t_k)+u(x,t_{k+1}))+\Phi_T^{k+1}\\ &\equiv \mathcal{J}_x^N u(x,t_{k+1})+\Phi_T^{k+1}\end{aligned} \tag{6}$$

where $\Phi_T^{k+1}=(\mathcal{J}_x^t-\mathcal{J}_x^N)(x,t_{k+1})$ denotes the remainder term. Substituting (4) and (6) into (1); yields the following system of semidiscrete problems for $k=0,1,...,N-1$.

$$\begin{cases}\left(\mathcal{D}_{L1}^{\alpha}-\dfrac{\partial^2}{\partial x^2}\right)u(x,t_{k+1})+\lambda\mathcal{J}_x^N u(x,t_{k+1})=R_{(1)\Delta t}^{k+1},\\ u(x,t_0)=\phi(x),\quad \text{for}\quad x\in\overline{\Omega}_x,\\ u(x,t_{k+1})=\psi(x,t_{k+1}),\quad \text{for}\quad x\in\partial\Omega_x.\end{cases} \tag{7}$$

The truncation error $R_{(1)\Delta t}^{k+1}=\Phi_{L1}^{k+1}+\Phi_T^{k+1}$ satisfies the following bound [45, 46]:

$$\left|R_{(1)\Delta t}^{k+1}\right|\le\left|\Phi_{L1}^{k+1}\right|+\left|\Phi_T^{k+1}\right|\le\mathcal{C}_u(\Delta t)^{2-\alpha}+\mathcal{C}_u(\Delta t)^{2-\beta}\le\mathcal{C}_u(\Delta t)^{2-\alpha}\quad\text{for}\quad k=0,1,\ldots,N-1, \tag{8}$$

provided $\beta\le\alpha$. Here $\mathcal{C}_u$ is a generic positive constant, which depends on the solution. The total error bound corresponding to the time semi-discretization will be proved later.

Remark 1 The bound given in (8) demonstrates that the truncation error bound $\mathcal{O}\left((\Delta t)^{2-\beta}\right)$ for the weakly singular integral operator [46, 47] is dominated by the truncation error bound of the L1 discretization of the fractional operator, which is of $\mathcal{O}\left((\Delta t)^{2-\alpha}\right)$ if $\beta\le\alpha$. In contrast, if $\beta\ge\alpha$, one can achieve the truncation error bound $\mathcal{O}\left((\Delta t)^{2-\beta}\right)$ for temporal semi-discretization.

Remark 2 If the kernel is smooth (instead of the present weakly singular nature), the discretization of the integral operator turns into the repeated quadrature rule through a trapezoidal approximation. It leads to the error bound $\mathcal{O}\left((\Delta t)^2\right)$ (see [48]). In this case, the truncation error bound of the fractional operator dominates the error bound of the integral operator, leading to the $\mathcal{O}\left((\Delta t)^{2-\alpha}\right)$ error bound.

Now let us consider the following finite difference scheme corresponding to equation (1):

$$\mathcal{D}_{L1}^{\alpha}U_{(1)}^{k+1}(x)=\frac{\partial^2 U_{(1)}^{k+1}(x)}{\partial x^2}-\lambda\mathcal{J}_x^N U_{(1)}^{k+1}(x),\quad k=0,1,\ldots,N-1, \tag{9}$$

where $U_{(1)}^{k+1}(x)$ is an approximation of $u\left(x,t_{k+1}\right)$. By omitting the dependence of $U_{(1)}^{k+1}(x)$ on x, the scheme (9) can be rewritten in the following form:

$$
\begin{aligned}
b_0 U_{(1)}^{k+1} - \Gamma(2-\alpha)(\Delta t)^{\alpha} \frac{\partial^2 U_{(1)}^{k+1}}{\partial x^2} &= b_0 U_{(1)}^{k} - \sum_{j=1}^{k} b_j \left[U_{(1)}^{k+1-j} - U_{(1)}^{k-j} \right] \\
&\quad - \frac{\lambda \Gamma(2-\alpha)(\Delta t)^{1+\alpha-\beta}}{2(1-\beta)} K(x, t_{k+1}) \left[U_{(1)}^{k+1} + U_{(1)}^{k} \right] \\
&\quad - \frac{\lambda \Gamma(2-\alpha)(\Delta t)^{1+\alpha}}{2} \sum_{j=0}^{k-1} \left[\mathbb{K}(x, t_{k+1}, t_j) U_{(1)}^{j} + \mathbb{K}(x, t_{k+1}, t_{j+1}) U_{(1)}^{j+1}) \right] \\
&= (b_0 - b_1) U_{(1)}^{k} + b_k U_{(1)}^{0} + \sum_{j=1}^{k-1} (b_j - b_{j+1}) U_{(1)}^{k-j} \\
&\quad - \frac{\lambda \Gamma(2-\alpha)(\Delta t)^{1+\alpha-\beta}}{2(1-\beta)} K(x, t_{k+1}) \left[U_{(1)}^{k+1} + U_{(1)}^{k} \right] \\
&\quad - \frac{\lambda \Gamma(2-\alpha)(\Delta t)^{1+\alpha}}{2} \sum_{j=0}^{k-1} \left[\mathbb{K}(x, t_{k+1}, t_j) U_{(1)}^{j} + \mathbb{K}(x, t_{k+1}, t_{j+1}) U_{(1)}^{j+1}) \right].
\end{aligned} \tag{10}
$$

It should be noted that the third term in the RHS of (10) vanishes when $k \leq 1$ and the last term vanishes for $k = 0$. It is also direct to check that

$$
\begin{cases}
b_j > 0, \quad j = 0, 1, \ldots, k, \\
1 = b_0 > b_1 > \cdots > b_k, \quad b_k \to 0 \quad \text{as } k \to \infty, \\
\displaystyle\sum_{j=0}^{k} (b_j - b_{j+1}) + b_{k+1} = (1 - b_1) + \sum_{j=1}^{k-1} (b_j - b_{j+1}) + b_k = 1.
\end{cases} \tag{11}
$$

Let us introduce the parameter α_0 as:

$$
\alpha_0 := \Gamma(2-\alpha)(\Delta t)^{\alpha}.
$$

The parameter α_0 is chosen purely as a notational simplification to streamline the presentation of the mathematical expressions that follow. This notation has also been used in a similar context in [45], where the authors used the L1 discretization on a uniform mesh in time to solve time-fractional PDEs without time singularities. However, the present research is based on the L1/L1-2 Legendre spectral method to solve the time-fractional integro-PDEs involving weakly singular kernels. We also extend the approach to solve more general integro-PDEs in two spatial dimensions (similar to [49, 50]) having initial time singularities in the numerical section. Note that $b_0 = 1$. Then, by reformulating the right-hand side of (10), we obtain an equivalent form of the semi-discrete time scheme (9) for $k \geq 2$:

$$
\widehat{R}_{(1)}^{k+1}(x) U_{(1)}^{k+1} - \alpha_0 \frac{\partial^2 U_{(1)}^{k+1}}{\partial x^2} = \widehat{F}_{(1)}^{k+1}(x), \tag{12}
$$

where

$$
\widehat{R}_{(1)}^{k+1}(x) = 1 + \frac{\lambda \alpha_0 (\Delta t)^{1-\beta}}{2(1-\beta)} K(x, t_{k+1}), \tag{13}
$$

and

$$\widehat{F}_{(1)}^{k+1}(x) = \left(1 - b_1 - \frac{\lambda\alpha_0(\Delta t)^{1-\beta}}{2(1-\beta)}K(x,t_{k+1})\right)U_{(1)}^{k} + b_k U_{(1)}^{0} + \sum_{j=1}^{k-1}(b_j - b_{j+1})U_{(1)}^{k-j} - \frac{\lambda\alpha_0\Delta t}{2}\sum_{j=0}^{k-1}\left[\mathbb{K}(x,t_{k+1},t_j)U_{(1)}^{j} + \mathbb{K}(x,t_{k+1},t_{j+1})U_{(1)}^{j+1})\right]. \tag{14}$$

In particular, when $k = 1$, the scheme (9) becomes

$$\left(1 + \frac{\lambda\alpha_0(\Delta t)^{1-\beta}}{2(1-\beta)}K(x,t_2)\right)U_{(1)}^{2} - \alpha_0\frac{\partial^2 U_{(1)}^{2}}{\partial x^2} = \left(1 - b_1 - \frac{\lambda\alpha_0(\Delta t)^{1-\beta}}{2(1-\beta)}K(x,t_2)\right)U_{(1)}^{1} + b_1 U_{(1)}^{0} - \frac{\lambda\alpha_0\Delta t}{2}\left[\mathbb{K}(x,t_2,t_0)U_{(1)}^{0} + \mathbb{K}(x,t_2,t_1)U_{(1)}^{1})\right]. \tag{15}$$

For the specific case when $k = 0$, it turns into

$$\left(1 + \frac{\lambda\alpha_0(\Delta t)^{1-\beta}}{2(1-\beta)}K(x,t_1)\right)U_{(1)}^{1} - \alpha_0\frac{\partial^2 U_{(1)}^{1}}{\partial x^2} = \left(1 - \frac{\lambda\alpha_0(\Delta t)^{1-\beta}}{2(1-\beta)}K(x,t_1)\right)U_{(1)}^{0}. \tag{16}$$

Hence, the expressions given in (12), (15) and (16) together with the initial and boundary conditions give the temporal semi-discretization of the proposed model (1). It will be useful to define the total semi-discretized error term $R_{(1)}^{k+1}$ by

$$R_{(1)}^{k+1} := \alpha_0\left[\frac{\partial^\alpha u(x,t_{k+1})}{\partial t^\alpha} - \mathcal{D}_{L1}^{\alpha}u(x,t_{k+1}) + \lambda\left(\mathcal{J}_x^t u(x,t_{k+1}) - \mathcal{J}_x^N u(x,t_{k+1})\right)\right]. \tag{17}$$

Then, we have from (8)

$$\left|R_{(1)}^{k+1}\right| = \alpha_0\left|R_{(1)\Delta t}^{k+1}\right| \leqslant \mathcal{C}_u(\Delta t)^2. \tag{18}$$

Note 1 It is important to note that the. truncation error bound given in (18) is valid only when $\beta \leq \alpha$. However, when $\beta > \alpha$, one has $\left|R_{(1)\Delta t}^{k+1}\right| \leq \mathcal{C}_u(\Delta t)^{2-\beta}$, as discussed in Remark 1. Consequently, the bound becomes $\mathcal{O}\left((\Delta t)^{2+\alpha-\beta}\right)$. Also, note that the bound given in (18) remains unchanged even if the kernel is considered to be sufficiently smooth. In this case, both the parameters α, β lie in the interval (0, 1) as already defined in the model problem.

In order to proceed further, assume the spatial domain $\Omega_x = (-L, L)$. The variational (weak) formulation of the Eq. (12) subject to the homogeneous boundary conditions $U_{(1)}^{k+1}(-L) = U_{(1)}^{k+1}(L) = 0$ for $k \geq 0$ reads: Find $U_{(1)}^{k+1} \in H_0^1(\Omega_x)$, such that

$$\left(\widehat{R}_{(1)}^{k+1}(x)U_{(1)}^{k+1},v\right)+\alpha_0\left(\frac{\partial U_{(1)}^{k+1}}{\partial x},\frac{\partial v}{\partial x}\right)=\left(\widehat{F}_{(1)}^{k+1}(x),v\right),\quad\forall v\in H_0^1(\Omega_x),\tag{19}$$

where $\widehat{R}_{(1)}^{k+1}(x)$ and $\widehat{F}_{(1)}^{k+1}(x)$ are defined in (13) and (14).

2.2 Semi-discretization using the L1-2 technique

Let $t_{j+\frac{1}{2}}=\frac{t_{j+1}+t_j}{2}$. For $k\geq 0$, we define

$$\begin{cases}\delta_t u_{j-\frac{1}{2}}=\dfrac{1}{\Delta t}\left[u\left(x,t_j\right)-u\left(x,t_{j-1}\right)\right],\\ \delta_t u_{j+\frac{1}{2}}=\dfrac{1}{\Delta t}\left[u\left(x,t_{j+1}\right)-u\left(x,t_j\right)\right],\\ \delta_t^2 u_j=\dfrac{1}{\Delta t}\left[\delta_t u_{j+\frac{1}{2}}-\delta_t u_{j-\frac{1}{2}}\right].\end{cases}$$

At the point $(\cdot,t_{k+1})$, where $t_{k+1}\in\Omega_t^N$, $\frac{\partial^\alpha u}{\partial t^\alpha}$ can be rewritten as:

$$\begin{aligned}\frac{\partial^\alpha u\left(x,t_{k+1}\right)}{\partial t^\alpha}&=\frac{1}{\Gamma(1-\alpha)}\left[\int_{s=t_0}^{t_1}\left(t_{k+1}-s\right)^{-\alpha}\frac{\partial u}{\partial s}(x,s)ds\right.\\&\left.+\sum_{j=1}^{k}\int_{s=t_j}^{t_{j+1}}\left(t_{k+1}-s\right)^{-\alpha}\frac{\partial u}{\partial s}(x,s)ds\right],\end{aligned}$$

which is approximated by

$$\begin{aligned}\frac{\partial^\alpha u\left(x,t_{k+1}\right)}{\partial t^\alpha}&=\frac{1}{\Gamma(1-\alpha)}\left[\int_{s=t_0}^{t_1}\frac{\frac{\partial}{\partial s}\prod\limits_{1,1}u(x,s)}{\left(t_{k+1}-s\right)^\alpha}ds+\sum_{j=1}^{k}\int_{s=t_j}^{t_{j+1}}\frac{\frac{\partial}{\partial s}\prod\limits_{2,j}u(x,s)}{\left(t_{k+1}-s\right)^\alpha}ds\right]+\Phi_{L12}^{k+1}\\&\equiv\mathcal{D}_{L12}^\alpha u\left(x,t_{k+1}\right)+\Phi_{L12}^{k+1},\end{aligned}\tag{20}$$

where Φ_{L12}^{k+1} denotes the truncation error for the discretization of fractional operators based on L1-2 method. The linear interpolation polynomial of u on $[t_{j-1},t_j]$ for $j\geq 1$ is

$$\prod_{1,j}u(x,t)=\frac{t_j-t}{\Delta t}u\left(x,t_{j-1}\right)+\frac{t-t_{j-1}}{\Delta t}u\left(x,t_j\right).\tag{21}$$

Moreover, the quadratic interpolation polynomial of u on $[t_j,t_{j+1}]$ for $j\geq 1$ is

$$\prod_{2,j} u(x,t) = \frac{(t-t_j)(t-t_{j+1})}{2(\Delta t)^2} u(x,t_{j-1}) + \frac{(t-t_{j-1})(t_{j+1}-t)}{(\Delta t)^2} u(x,t_j) + \frac{(t-t_j)(t-t_{j-1})}{2(\Delta t)^2} u(x,t_{j+1}). \tag{22}$$

The L1-2 scheme [51, 52] is produced by replacing (21) and (22) in (20) as:

$$\begin{aligned}
\mathcal{D}^{\alpha}_{L12} u(x,t_{k+1}) =& \frac{1}{\Gamma(1-\alpha)} \left[\delta_t u_{\frac{1}{2}} \int_{s=t_0}^{t_1} \frac{ds}{(t_{k+1}-s)^{\alpha}} \right. \\
&\left. + \sum_{j=1}^{k} \int_{s=t_j}^{t_{j+1}} \frac{\delta_t u_{j+\frac{1}{2}} + \left(s - t_{j+\frac{1}{2}}\right) \delta_t^2 u_j}{(t_{k+1}-s)^{\alpha}} ds \right] \\
=& \frac{1}{\Gamma(1-\alpha)} \left[\sum_{j=0}^{k} \delta_t u_{j+\frac{1}{2}} \int_{s=t_j}^{t_{j+1}} \frac{ds}{(t_{k+1}-s)^{\alpha}} \right. \\
&\left. + \sum_{j=1}^{k} \delta_t^2 u_j \int_{s=t_j}^{t_{j+1}} \frac{s - t_{j+\frac{1}{2}}}{(t_{k+1}-s)^{\alpha}} ds \right] \\
=& \frac{(\Delta t)^{-\alpha}}{\Gamma(2-\alpha)} \left[\sum_{j=0}^{k} \{u(x,t_{j+1}) - u(x,t_j)\} b_{k-j} \right. \\
&\left. + \sum_{j=1}^{k} \{u(x,t_{j+1}) - 2u(x,t_j) + u(x,t_{j-1})\} b^*_{k-j} \right] \\
=& \frac{(\Delta t)^{-\alpha}}{\Gamma(2-\alpha)} \sum_{j=0}^{k} [u(x,t_{k+1-j}) - u(x,t_{k-j})] d_{k-j}.
\end{aligned} \tag{23}$$

The coefficients are $d_0 = 1$ for $k = 0$, and for $k \geq 1$

$$d_j = \begin{cases} b_j + b^*_j, & \text{for } j = 0, \\ b_j + b^*_j - b^*_{j-1}, & \text{for } 2 \leq j \leq k-1, \\ b_j - b^*_{j-1}, & \text{for } j = k, \end{cases}$$

where b_j is defined in (5) and b^*_j are defined as

$$b^*_j = \frac{1}{2-\alpha} \left[(j+1)^{2-\alpha} - j^{2-\alpha} \right] - \frac{1}{2} \left[(j+1)^{1-\alpha} + j^{1-\alpha} \right], \quad \text{for } j = 0, 2, \ldots, N-1. \tag{24}$$

Upon replacing (23) and (6) in (1), we get to the following system

$$\begin{cases} \left(\mathcal{D}^{\alpha}_{L12} - \dfrac{\partial^2}{\partial x^2}\right) u\left(x, t_{k+1}\right) + \lambda \mathcal{J}^{N}_{x} u\left(x, t_{k+1}\right) = R^{k+1}_{(2)\Delta t}, \\ u(x, t_0) = \phi(x), \quad \text{for} \quad x \in \overline{\Omega}_x, \\ u(x, t_{k+1}) = \psi(x, t_{k+1}), \quad \text{for} \quad x \in \partial\Omega_x, \end{cases} \tag{25}$$

with $R^{k+1}_{(2)\Delta t} = \Phi^{k+1}_{L12} + \Phi^{k+1}_{T}$ for every $k = 0, 1, 2, \ldots, N-1$. We define the discrete fractional differential operator $\mathcal{D}^{\alpha}_{L12}$ as

$$\mathcal{D}^{\alpha}_{L12} u(x, t_{k+1}) := \frac{1}{\Gamma(2-\alpha)} \sum_{j=0}^{k} d_j \frac{u\left(x, t_{k+1-j}\right) - u\left(x, t_{k-j}\right)}{\Delta t^{\alpha}} + \Phi^{k+1}_{L12}.$$

The truncation error $R^{(k+1)}_{(2)\Delta t}$ [46, 48] satisfies the following bound:

$$\left|R^{k+1}_{(2)\Delta t}\right| \leq \begin{cases} \mathcal{C}_u(\Delta t)^{2-\alpha} + \mathcal{C}_u(\Delta t)^{2-\beta} \leq \mathcal{C}_u(\Delta t)^{2-\alpha} & \text{for} \quad k = 0, \\ \mathcal{C}_u(\Delta t)^{3-\alpha} + \mathcal{C}_u(\Delta t)^{2-\beta} \leq \mathcal{C}_u(\Delta t)^{2-\beta} & \text{for} \quad k = 1, 2, \ldots, N-1, \end{cases} \tag{26}$$

provided $\beta \leq \alpha$.

Remark 3 For $\beta \geq \alpha$, it can be noticed that the bound given in (26) becomes $\left|R^{k+1}_{(2)\Delta t}\right| \leq \mathcal{C}_u(\Delta t)^{2-\beta}$ for all $k = 0, 1, \ldots, N-1$. In the case of a smooth kernel, the bounds can be expressed as:

$$\left|R^{k+1}_{(2)\Delta t}\right| \leq \begin{cases} \mathcal{C}_u(\Delta t)^{2-\alpha} & \text{for} \quad k = 0, \\ \mathcal{C}_u(\Delta t)^{2} & \text{for} \quad k = 1, 2, \ldots, N-1. \end{cases}$$

Remark 4 The L1-2 discretization, while requiring marginally more operations per time step, achieves higher-order accuracy $\mathcal{O}((\Delta t)^{3-\alpha})$ in time, leading to improved overall computational efficiency for a given error tolerance. See Table 14 for Example 6.2 in the numerical experiment section for efficiency and accuracy comparisons.

Using $\mathcal{D}^{\alpha}_{L12} u(x, t_{k+1})$ as an approximation of $\frac{\partial^{\alpha} u(x, t_{k+1})}{\partial t^{\alpha}}$, let us state the following time semi-discretization scheme based on L1-2 scheme for (1):

$$\mathcal{D}^{\alpha}_{L12} U^{k+1}_{(2)}(x) = \frac{\partial^2 U^{k+1}_{(2)}(x)}{\partial x^2} - \lambda \mathcal{J}^{N}_{x} U^{k+1}_{(2)}(x), \quad k = 0, 1, \ldots, N-1, \tag{27}$$

which can be simplified as:

$$
\begin{aligned}
d_0 U_{(2)}^{k+1} - \Gamma(2-\alpha)(\Delta t)^{\alpha}\frac{\partial^2 U_{(2)}^{k+1}}{\partial x^2} &= d_0 U_{(2)}^{k} - \sum_{j=1}^{k} d_j \left[U_{(2)}^{k+1-j} - U_{(2)}^{k-j}\right] \\
&\quad - \frac{\lambda\Gamma(2-\alpha)(\Delta t)^{1+\alpha-\beta}}{2(1-\beta)} K(x,t_{k+1})\left[U_{(2)}^{k+1} + U_{(2)}^{k}\right] \\
&\quad - \frac{\lambda\Gamma(2-\alpha)(\Delta t)^{1+\alpha}}{2}\sum_{j=0}^{k-1}\left[\mathbb{K}(x,t_{k+1},t_j)U_{(2)}^{j} + \mathbb{K}(x,t_{k+1},t_{j+1})U_{(2)}^{j+1})\right] \\
&= (d_0 - d_1)U_{(2)}^{k} + d_k U_{(2)}^{0} + \sum_{j=1}^{k-1}(d_j - d_{j+1})U_{(2)}^{k-j} \\
&\quad - \frac{\lambda\Gamma(2-\alpha)(\Delta t)^{1+\alpha-\beta}}{2(1-\beta)} K(x,t_{k+1})\left[U_{(2)}^{k+1} + U_{(2)}^{k}\right] \\
&\quad - \frac{\lambda\Gamma(2-\alpha)(\Delta t)^{1+\alpha}}{2}\sum_{j=0}^{k-1}\left[\mathbb{K}(x,t_{k+1},t_j)U_{(2)}^{j} + \mathbb{K}(x,t_{k+1},t_{j+1})U_{(2)}^{j+1})\right].
\end{aligned}
\tag{28}
$$

It is worth noting that the third term in the RHS of (28) vanishes when $k \leqslant 1$, while the last term of (28) vanishes when $k = 0$. It is direct to check that

$$
\begin{cases}
d_j > 0, \quad j = 0, 1, \ldots, k, \\
1 = d_0 > d_1 > \cdots > d_k, \quad d_k \to 0 \quad \text{as } k \to \infty \\
\displaystyle\sum_{j=0}^{k}(d_j - d_{j+1}) + d_{k+1} = (1 - d_1) + \sum_{j=1}^{k-1}(d_j - d_{j+1}) + d_k = 1.
\end{cases}
\tag{29}
$$

Let us introduce the parameter α_0 as $\alpha_0 := \Gamma(2-\alpha)\Delta t^{\alpha}$. Note that $d_0 = 1$, and then by reformulating the right-hand side of (28), we obtain an equivalent form to the scheme (27):

$$
\widehat{R}_{(2)}^{k+1}(x)U_{(2)}^{k+1} - \alpha_0\frac{\partial^2 U_{(2)}^{k+1}}{\partial x^2} = \widehat{F}_{(2)}^{k+1}(x),
\tag{30}
$$

where

$$
\widehat{R}_{(2)}^{k+1}(x) = 1 + \frac{\lambda\alpha_0(\Delta t)^{1-\beta}}{2(1-\beta)}K(x,t_{k+1}),
\tag{31}
$$

and

$$
\begin{aligned}
\widehat{F}_{(2)}^{k+1}(x) &= \left(1 - d_1 - \frac{\lambda\alpha_0(\Delta t)^{1-\beta}}{2(1-\beta)}K(x,t_{k+1})\right)U_{(2)}^{k} + d_k U_{(2)}^{0} + \sum_{j=1}^{k-1}(d_j - d_{j+1})U_{(2)}^{k-j} \\
&\quad - \frac{\lambda\alpha_0\Delta t}{2}\sum_{j=0}^{k-1}\left[\mathbb{K}(x,t_{k+1},t_j)U_{(2)}^{j} + \mathbb{K}(x,t_{k+1},t_{j+1})U_{(2)}^{j+1})\right].
\end{aligned}
\tag{32}
$$

In particular, when $k = 1$, scheme (30) becomes

$$\left(1+\frac{\lambda\alpha_0(\Delta t)^{1-\beta}}{2(1-\beta)}K(x,t_2)\right)U^2_{(2)}-\alpha_0\frac{\partial^2U^2_{(2)}}{\partial x^2}$$
$$=\left(1-d_1-\frac{\lambda\alpha_0(\Delta t)^{1-\beta}}{2(1-\beta)}K(x,t_2)\right)U^1_{(2)}+d_1U^0_{(2)} \tag{33}$$
$$-\frac{\lambda\alpha_0\Delta t}{2}\left[\mathbb{K}(x,t_2,t_0)U^0_{(2)}+\mathbb{K}(x,t_2,t_1)U^1_{(2)})\right].$$

For the specific case when $k=0$, it becomes

$$\left(1+\frac{\lambda\alpha_0(\Delta t)^{1-\beta}}{2(1-\beta)}K(x,t_1)\right)U^1_{(2)}-\alpha_0\frac{\partial^2U^1_{(2)}}{\partial x^2}=\left(1-\frac{\lambda\alpha_0(\Delta t)^{1-\beta}}{2(1-\beta)}K(x,t_1)\right)U^0_{(2)}. \tag{34}$$

Hence, the expressions given in (30), (33) and (34) together with the initial and boundary conditions give the temporal semi-discretization of the proposed model (1) based on the L1-2 method. It will be useful to define the error term $R^{k+1}_{(2)}$ by

$$R^{k+1}_{(2)}:=\alpha_0\left[\frac{\partial^\alpha u\left(x,t_{k+1}\right)}{\partial t^\alpha}-\mathcal{D}^\alpha_{L12}u\left(x,t_{k+1}\right)+\lambda\left(\mathcal{J}^t_xu(x,t_{k+1})-\mathcal{J}^N_xu(x,t_{k+1})\right)\right]. \tag{35}$$

Then we have from (26)

$$\left|R^{k+1}_{(2)}\right|=\Gamma(2-\alpha)\Delta t^\alpha\left|R^{k+1}_{(2)\Delta t}\right|\leq\begin{cases}\mathcal{C}_u(\Delta t)^2 & \text{for } k=0,\\ \mathcal{C}_u(\Delta t)^{2+\alpha-\beta} & \text{for } k=1,2,\ldots,N-1,\end{cases} \tag{36}$$

Note 2 It is important to note that the bound given in (36) is valid only when $\beta\leq\alpha$. However, when $\beta\geq\alpha$, as per the discussion in Remark 3, the bound will be $\mathcal{O}\left((\Delta t)^{2+\alpha-\beta}\right)$ for all $k=0,1,\ldots,N-1$. In contrast, for a sufficiently smooth kernel, the bound becomes:

$$\left|R^{k+1}_{(2)}\right|\leq\begin{cases}\mathcal{C}_u(\Delta t)^2 & \text{for } k=0,\\ \mathcal{C}_u(\Delta t)^{2+\alpha} & \text{for } k=1,2,\ldots,N-1,\end{cases}.$$

The variational (weak) formulation of the Eq.(30) subject to the boundary conditions $U^{k+1}_{(2)}(-L)=U^{k+1}_{(2)}(L)=0$ for $k\geq0$ reads: find $U^{k+1}_{(2)}\in H^1_0(\Omega_x)$, such that

$$\left(\widehat{R}^{k+1}_{(2)}U^{k+1}_{(2)},v\right)+\alpha_0\left(\frac{\partial U^{k+1}_{(2)}}{\partial x},\frac{\partial v}{\partial x}\right)=\left(\widehat{F}^{k+1}_{(2)},v\right),\forall v\in H^1_0(\Omega_x), \tag{37}$$

where $\widehat{R}^{k+1}_{(2)}(x)$ and $\widehat{F}^{k+1}_{(2)}(x)$ are defined in (31) and (32).

3 Stability and convergence of the temporal semi-discretization

Now we consider the stability of the above approximation. The following stability result applies to the semi-discretized problem.

Theorem 3.1 *The semi-discretized problems (19) and (37) are stable for a sufficiently large number of partitions in time. That is, there exists one $\Delta t > 0$, such that*

$$\left\| U_{(q)}^{k+1} \right\|_1 \leqslant \left\| U_{(q)}^{0} \right\|, \quad q = 1, 2 \text{ and } k = 0, 1, \ldots, N-1.$$

Proof We will prove the result by induction. First, let us take $k = 0$. Then, the weak formulation given in (19) and (37), becomes

$$\left(\widehat{R}_{(q)}^{1} U_{(q)}^{1}, v \right) + \alpha_0 \left(\frac{\partial U_{(q)}^{1}}{\partial x}, \frac{\partial v}{\partial x} \right) = \left(\widehat{F}_{(q)}^{1}, v \right) \quad \forall v \in H_0^1(\Omega_x), \ q = 1, 2,$$

where $\widehat{R}_{(q)}^{1}(x) = \left(1 + \frac{\lambda \alpha_0 (\Delta t)^{1-\beta}}{2(1-\beta)} K(x, t_1) \right)$ and $\widehat{F}_{(q)}^{1}(x) = \left(1 - \frac{\lambda \alpha_0 (\Delta t)^{1-\beta}}{2(1-\beta)} K(x, t_1) \right) U_{(q)}^{0}$. Note that $K > 0$, which yields $\widehat{R}_{(q)}^{1} > 1$. In addition, K is sufficiently smooth, $\alpha_0, \Delta t$ is bounded, and $(1 - \beta) > 0$. Therefore, $\left| 1 - \frac{\lambda \alpha_0 (\Delta t)^{1-\beta}}{2(1-\beta)} K \right| < 1$ for a sufficiently large number of mesh points in the temporal direction. Note that $\|v\| \leqslant \|v\|_1$. Then, by using $v = U_{(q)}^{1}$ in the above equation, along with the Cauchy-Schwarz inequality, it leads to the following:

$$\left\| U_{(q)}^{1} \right\|_1^2 \leq \left(\widehat{R}_{(q)}^{1} U_{(q)}^{1}, U_{(q)}^{1} \right) + \alpha_0 \left(\frac{\partial U_{(q)}^{1}}{\partial x}, \frac{\partial U_{(q)}^{1}}{\partial x} \right) \leqslant \left\| U_{(q)}^{0} \right\| \left\| U_{(q)}^{1} \right\|.$$

Consequently, we have $\left\| U_{(q)}^{1} \right\|_1 \leq \left\| U_{(q)}^{0} \right\|$. Now, suppose it is true that

$$\left\| U_{(q)}^{j} \right\|_1 \leqslant \left\| U_{(q)}^{0} \right\|, \quad j = 2, 3, \ldots, k. \tag{38}$$

We want to prove $\left\| U_{(q)}^{k+1} \right\|_1 \leqslant \left\| U_{(q)}^{0} \right\|$ for $q = 1, 2$. Let us prove the case for $q = 1$. Taking $v = U_{(1)}^{k+1}$ in (19), it gives

$$\left(\widehat{R}_{(1)}^{k+1} U_{(1)}^{k+1}, U_{(1)}^{k+1} \right) + \alpha_0 \left(\frac{\partial U_{(1)}^{k+1}}{\partial x}, \frac{\partial U_{(1)}^{k+1}}{\partial x} \right) = \left(\widehat{F}_{(1)}^{k+1}(x), U_{(1)}^{k+1} \right). \tag{39}$$

Observe that the kernel function $K > 0$, which yields $\widehat{R}_{(1)}^{k+1} > 1$. Consequently, applying the Cauchy-Schwarz inequality to (39), it follows:

$$\left\|U_{(1)}^{k+1}\right\|_1^2 \leq \left\|\widehat{F}_{(1)}^{k+1}(x)\right\| \left\|U_{(1)}^{k+1}\right\| \leq \left\|\widehat{F}_{(1)}^{k+1}(x)\right\| \left\|U_{(1)}^{k+1}\right\|_1. \tag{40}$$

From (14) and (38), we have

$$\left\|\widehat{F}_{(1)}^{k+1}(x)\right\| \leq \mathcal{C}_x \left\|U_{(1)}^{0}\right\|.$$

where $\mathcal{C}_x = |1 - b_1 - \frac{\lambda\alpha_0(\Delta t)^{1-\beta}}{2(1-\beta)}K(x,t_{k+1}) + b_k + \sum_{j=1}^{k-1}(b_j - b_{j+1}) - \frac{\lambda\alpha_0\Delta t}{2}$ $\sum_{j=0}^{k-1}[\mathbb{K}(x,t_{k+1},t_j)+\mathbb{K}(x,t_{k+1},t_{j+1})]|$. Note that $K \geq 0$, is bounded and $\mathbb{K} \geq 0$. In particular, the last term inside the above modulus goes to zero since $\alpha_0 \to 0$ whenever $\Delta t \to 0$. Therefore, using the last inequality of (11), there exists a constant $\mathcal{C} < 1$, such that $\mathcal{C}_x < \mathcal{C}$. Consequently, (40) yields

$$\left\|U_{(1)}^{k+1}\right\|_1 \leqslant \left\|U_{(1)}^{0}\right\|. \tag{41}$$

Therefore, the inequality (38) is true for $j = k+1$. Hence, by the principle of mathematical induction, the theorem is proved for $q = 1$. Similarly, using the last inequality of (29), this theorem can be proved for $q = 2$. □

To investigate the precision of our method, we now present a detailed error analysis of temporal semi-discretization based on the L1/L1-2 schemes in the context of weakly singular time-fractional integro-PDEs (1).

Theorem 3.2 *Let u be the exact solution of (1). $\left\{U_{(q)}^{k}\right\}_{k=0}^{N}$ be the discrete time solution of (19) and (37) with the initial condition $U_{(q)}^{0}(x) = u(x,0)$ for $q = 1, 2$, respectively. Then we have the following error estimates:*

$$\left\|u(t_k) - U_{(q)}^{k}\right\|_1 \leq \begin{cases} \mathcal{C}_{u,\alpha}N^{-(2-\alpha)}, & q = 1,\ k = 1,2,\ldots,N. \\ \mathcal{C}_{u,\alpha}N^{-(2-\beta)}, & q = 2,\ k = 2,\ldots,N, \\ \mathcal{C}_{u,\alpha}N^{-(2-\alpha)}, & q = 2,\ k = 1, \end{cases} \tag{42}$$

where $\mathcal{C}_{u,\alpha} := \dfrac{\mathcal{C}_u T^2}{1-\alpha}$, with $0 < \alpha < 1$.

Proof We start by proving the following estimate when $q = 1$:

$$\left\|u(t_j) - U_{(1)}^{j}\right\|_1 \leqslant \mathcal{C}_u b_{j-1}^{-1}(\Delta t)^2, \quad j = 1,2,\ldots,N. \tag{43}$$

As in the proof of Theorem (3.1), we will use the mathematical induction. Let $\bar{e}^k_{(1)} = u(x,t_k) - U^k_{(1)}(x)$. For $j=1$, by combining (1), (16) and (17), we have the following error equation in the weak form:

$$\left(\widehat{R}^1_{(1)}\bar{e}^1_{(1)}, v\right) + \alpha_0\left(\frac{\partial \bar{e}^1_{(1)}}{\partial x}, \frac{\partial v}{\partial x}\right) = \left(\left(1 - \frac{\lambda\alpha_0(\Delta t)^{1-\beta}}{2(1-\beta)}K(x,t_1)\right)\bar{e}^0_{(1)}, v\right) + \left(R^1_{(1)}, v\right), \ \forall v \in H^1_0(\Omega_x).$$

As we have, $\widehat{R}^1_{(1)} > 1$ and $\bar{e}^0_{(1)} = 0$, the Cauchy-Schwarz inequality gives the following for $v = \bar{e}^1_{(1)}$.

$$\left\|\bar{e}^1_{(1)}\right\|^2_1 \leqslant \left(\widehat{R}^1_{(1)}\bar{e}^1_{(1)}, v\right) + \alpha_0\left(\frac{\partial \bar{e}^1_{(1)}}{\partial x}, \frac{\partial v}{\partial x}\right) \leq \left\|R^1_{(1)}\right\|\left\|\bar{e}^1_{(1)}\right\|.$$

This, together with (18), gives

$$\left\|u(t_1) - U^1_{(1)}\right\|_1 \leqslant \mathcal{C}_u b_0^{-1}(\Delta t)^2,$$

where $b_0 = 1$. Therefore, (43) is true for the case $j=1$. Now, suppose that (43) holds for $j = 1, 2, \ldots, k$. We need to prove that it also holds for $j = k+1$. By combining (1), (17) and (19), we have the following error equation in the weak form:

$$\begin{aligned}\left(\widehat{R}^{k+1}_{(1)}\bar{e}^{k+1}_{(1)}, v\right) + \alpha_0\left(\frac{\partial \bar{e}^{k+1}_{(1)}}{\partial x}, \frac{\partial v}{\partial x}\right) &= \left(\left(1 - b_1 - \frac{\lambda\alpha_0(\Delta t)^{1-\beta}}{2(1-\beta)}K(x,t_{k+1})\right)\bar{e}^k_{(1)}, v\right) \\ &+ \sum_{j=1}^{k-1}(b_j - b_{j+1})\left(\bar{e}^{k-j}_{(1)}, v\right) \\ &+ b_k\left(\bar{e}^0_{(1)}, v\right) + \left(R^{k+1}_{(1)}, v\right) \\ &- \frac{\lambda\alpha_0\Delta t}{2}\sum_{j=0}^{k-1}\left[\mathbb{K}(x,t_{k+1},t_j)\bar{e}^j_{(1)} + \mathbb{K}(x,t_{k+1},t_{j+1})\bar{e}^{j+1}_{(1)}\right],\end{aligned} \tag{44}$$

for all $v \in H^1_0(\Omega_x)$. Note that $\bar{e}^0_{(1)} = 0$. Now, applying Cauchy-Schwarz inequality and taking $v = \bar{e}^{k+1}_{(1)}$ in (44), we get

$$\begin{aligned}\left\|\bar{e}_{(1)}^{k+1}\right\|_1^2 &\leqslant \left(\widehat{R}_{(1)}^{k+1}\bar{e}_{(1)}^{k+1}, v\right) + \alpha_0\left(\frac{\partial \bar{e}_{(1)}^{k+1}}{\partial x}, \frac{\partial v}{\partial x}\right)\\ &\leqslant \left(1 - b_1 - \frac{\lambda\alpha_0(\Delta t)^{1-\beta}}{2(1-\beta)}K(x,t_{k+1})\right)\left\|\bar{e}_{(1)}^{k}\right\|\left\|\bar{e}_{(1)}^{k+1}\right\|\\ &\quad + \sum_{j=1}^{k-1}(b_j - b_{j+1})\left\|\bar{e}_{(1)}^{k-j}\right\|\left\|\bar{e}_{(1)}^{k+1}\right\| + \left\|R_{(1)}^{k+1}\right\|\left\|\bar{e}_{(1)}^{k+1}\right\|\\ &\quad - \frac{\lambda\alpha_0\Delta t}{2}\sum_{j=0}^{k-1}\left[\mathbb{K}(x,t_{k+1},t_j)\|\bar{e}_{(1)}^{j}\|\right.\\ &\quad \left. + \mathbb{K}(x,t_{k+1},t_{j+1})\|\bar{e}_{(1)}^{j+1}\|\right]\left\|\bar{e}_{(1)}^{k+1}\right\|.\end{aligned}$$

Dividing both sides by $\left\|\bar{e}_{(1)}^{k+1}\right\|_1$ and using the induction hypothesis, we obtain

$$\begin{aligned}\left\|\bar{e}_{(1)}^{k+1}\right\|_1 \leqslant \Bigg[&\left(1 - b_1 - \frac{\lambda\alpha_0(\Delta t)^{1-\beta}}{2(1-\beta)}K(x,t_{k+1})\right)b_{k-1}^{-1} + \sum_{j=1}^{k-1}(b_j - b_{j+1})\,b_{k-j-1}^{-1}\\ &- \frac{\lambda\alpha_0\Delta t}{2}\sum_{j=0}^{k-1}\left[\mathbb{K}(x,t_{k+1},t_j)b_{j-1}^{-1} + \mathbb{K}(x,t_{k+1},t_{j+1})b_j^{-1}\right]\Bigg]\mathcal{C}_u(\Delta t)^2 + \mathcal{C}_u(\Delta t)^2.\end{aligned}$$

Note that $\mathbb{K} > 0$, and $b_j > 0$ with the fact that $\frac{b_j^{-1}}{b_{j+1}^{-1}} < 1$ for all non-negative integers j. Also, we have $K > 0$, which is bounded. Then for a sufficiently large number of mesh points in the temporal direction, one has the following:

$$\left\|\bar{e}_{(1)}^{k+1}\right\|_1 \leqslant \left[(1 - b_1) + \sum_{j=1}^{k-1}(b_j - b_{j+1}) + b_k\right]\mathcal{C}_u b_k^{-1}(\Delta t)^2.$$

Using the last inequality of (11) in the above inequality; gives

$$\left\|\bar{e}_{(1)}^{k+1}\right\|_1 \leqslant \mathcal{C}_u b_k^{-1}(\Delta t)^2.$$

Thus the estimate given in (43) is proved. Now, by the definition of b_k, a direct computation shows that

$$k^{-\alpha}b_{k-1}^{-1} = \frac{k^{-\alpha}}{k^{1-\alpha} - (k-1)^{1-\alpha}} \to \frac{1}{1-\alpha}, \qquad \text{as } k \to \infty,$$

and the function $\phi(x) := \frac{x^{-\alpha}}{x^{1-\alpha} - (x-1)^{1-\alpha}}$ is increasing on x for all $x > 1$ as

$$\phi'(x) = \frac{1}{x(x-1)^{\alpha}}\left[1-\left(1-\frac{1}{x}\right)^{\alpha}-\frac{\alpha}{x}\right] \geqslant 0 \quad \forall x > 1, \quad 0 \leqslant \alpha \leqslant 1.$$

This means $k^{-\alpha}b_{k-1}^{-1}$ is increasingly tending to $\frac{1}{1-\alpha}$ as $1 < k \to \infty$. Note that $k^{-\alpha}b_{k-1}^{-1} = 1$ when $k = 1$. Hence we have

$$k^{-\alpha}b_{k-1}^{-1} \leqslant \frac{1}{1-\alpha}, \quad k = 1, 2, \ldots, N. \tag{45}$$

Consequently, we obtain, for all k (satisfying $k\Delta t \leqslant T$), that

$$\begin{aligned}\left\|u\left(t_k\right) - U_{(1)}^{k}\right\|_1 &\leqslant \mathcal{C}_u b_{k-1}^{-1}(\Delta t)^2 = \mathcal{C}_u k^{-\alpha} b_{k-1}^{-1} k^{\alpha}(\Delta t)^2 \leqslant \frac{\mathcal{C}_u}{1-\alpha}(k\Delta t)^{\alpha}(\Delta t)^{2-\alpha} \\ &\leqslant \frac{\mathcal{C}_u}{1-\alpha}T^{\alpha}(\Delta t)^{2-\alpha} \leqslant \mathcal{C}_{u,\alpha}N^{-(2-\alpha)}.\end{aligned}$$

Thus, the theorem is proved for $q = 1$. We can also conclude that

$$\left\|u\left(t_k\right) - U_{(q)}^{k}\right\|_1 \leq \begin{cases} \mathcal{C}_{u,\alpha}N^{-(2-\beta)}, & q = 2, \ k = 2, \ldots, N, \\ \mathcal{C}_{u,\alpha}N^{-(2-\alpha)}, & q = 2, \ k = 1, \end{cases}$$

is true by carrying out the same process in a similar manner when $q = 2$. This completes our proof. □

Remark 5 Theorem 3.2 demonstrates that the temporal semi-discretization yields distinct convergence rates for the fractional and weakly singular integral operators. When using the L1 method ($q = 1$), the discretization error shows a convergence rate of $\mathcal{O}\left(N^{-(2-\alpha)}\right)$ for all $k \geq 1$. In contrast, the L1-2 method ($q = 2$) achieves a convergence rate of $\mathcal{O}\left(N^{-(2-\beta)}\right)$ when $k \geq 2$, while maintaining the convergence of $\mathcal{O}\left(N^{-(2-\alpha)}\right)$ for $k = 1$, provided $\beta \leq \alpha$. Furthermore, when $\beta > \alpha$, both of these schemes will lead to a convergent solution $\mathcal{O}\left(N^{-(2-\beta)}\right)$ in time, for all $k \geq 1$.

Remark 6 If one has a sufficiently smooth kernel for the integral operator, the error bounds presented in (42) can be refined as follows:

$$\left\|u\left(t_k\right) - U_{(q)}^{k}\right\|_1 \leq \begin{cases} \mathcal{C}_{u,\alpha}N^{-(2-\alpha)}, & q = 1, \ k = 1, 2, \ldots, N. \\ \mathcal{C}_{u,\alpha}N^{-2}, & q = 2, \ k = 2, \ldots, N, \\ \mathcal{C}_{u,\alpha}N^{-(2-\alpha)}, & q = 2, \ k = 1, \end{cases} \tag{46}$$

Remark 7 When $\lambda = 0$, the proposed problem (1) reduces to a time-fractional partial differential equation. For this special case, while the traditional L1-Legendre spectral method achieves a temporal accuracy of $\mathcal{O}(N^{-(2-\alpha)})$ as proved in [45], our pro-

posed L1-2-Legendre spectral method demonstrates superior performance. Specifically, it achieves an improved accuracy of $\mathcal{O}(N^{-(3-\alpha)})$ for all time steps t_n where $n = 2, 3, \ldots, N$, although it maintains the standard accuracy $\mathcal{O}(N^{-(2-\alpha)})$ in the first time step t_1. This theoretical convergence can also be confirmed in the numerical experiment section (see Example 6.2).

4 Legendre-Spectral method for spatial discretization

In this section, we introduce the Legendre spectral method, which is based on approximating integrals using Legendre Gauss-type quadratures. This method is particularly useful in solving differential equations as; it allows accurate numerical integration by leveraging orthogonal polynomials and their associated quadrature points. We need more notation to define this method. From now on, we consider $\Omega_x = (-1, 1)$. Let $L_M(x)$ represent the Legendre polynomial of degree M defined in Ω_x. The points $\xi_j, j = 0, 1, \ldots, M$, are referred; as the Legendre-Gauss-Lobatto (GLL) points, which are the zeros of the polynomial $\left(1 - x^2\right) L'_M(x); j = 0, 1, \ldots, M$. These points are nodes, where the function is evaluated for the spectral method. Let us define ω_j as the weights of the integral approximation. The corresponding Legendre collocation points $\xi_j, j = 0, 1, \ldots, M$, are chosen such that they satisfy the Legendre-Gauss-Lobatto quadrature formula. This quadrature approximates integrals of the form

$$\int_{-1}^{1} \Phi(x)\mathrm{d}x = \sum_{j=0}^{M} \Phi\left(\xi_j\right) \omega_j, \quad \forall \Phi(x) \in \mathbb{P}_{2M-1}(\Omega_x), \tag{47}$$

where $\mathbb{P}_{2M-1}(\Omega_x)$ is the space of all polynomials on Ω_x of degree less than or equal to $2M - 1$. Now let us define the discrete inner product approximation corresponding to any continuous functions θ and Υ, as

$$(\theta, \Upsilon)_M = \sum_{i=0}^{M} \theta\left(\xi_i\right) \Upsilon\left(\xi_i\right) \omega_i,$$

and the associated discrete norm $\|\theta\|_M := (\theta, \theta)_M^{1/2}$. It is well known that the discrete norm $\|\cdot\|_M$ is equivalent to the standard L^2-norm in $\mathbb{P}_M(\Omega_x)$:

$$\|\theta\|_0 \leqslant \|\theta\|_M \leqslant \sqrt{3}\|\theta\|_0, \quad \forall \theta \in \mathbb{P}_M(\Omega_x). \tag{48}$$

Let I_M be the interpolation operator based on the $M + 1$ GLL points, that is, $\forall \Upsilon \in C^0(\overline{\Omega}_x), I_M \Upsilon \in \mathbb{P}_M(\Omega_x)$, such that $I_M \Upsilon\left(\xi_j\right) = \Upsilon\left(\xi_j\right), j = 0, \ldots, M$. Define $P_M^0(\Omega_x) = P_M(\Omega_x) \cap H_0^1(\Omega_x)$. Now we consider the Legendre spectral approximation as follows: Find $U_{(1)M}^{k+1} \in P_M^0(\Omega_x)$, such that

$$B_M\left(U_{(1)M}^{k+1}, v_M\right) = F_{(1)}^{k+1,M}\left(v_M\right), \quad \text{for all } v_M \in \mathbb{P}_M^0(\Omega_x), \tag{49}$$

where the bilinear form $B_M(\cdot,\cdot)$ is defined by

$$B_M\left(U_{(1)M}^{k+1}, v_M\right) := \left(\widehat{R}_{(1)}^{k+1}(x)U_{(1)M}^{k+1}, v_M\right)_M + \alpha_0\left(\frac{\partial}{\partial x}U_{(1)M}^{k+1}, \frac{\partial}{\partial x}v_M\right)_M,$$

and the functional $F_{(1)}^{k+1,M}(\cdot)$ is given by

$$F_{(1)}^{k+1,M}\left(v_M\right) := \left(f(x,t_{k+1}), v_M\right)_M + \left(\widehat{F}_{(1)}^{k+1}(x), v_M\right)_M,$$

where $\widehat{R}_{(1)}^{k+1}(x)$ and $\widehat{F}_{(1)}^{k+1}(x)$ are defined in (13) and (14), respectively. Taking into account the norm equivalence (48), the well-posedness of the problem (49) is immediate with the help of the Lax-Milgram lemma. However, rigorous error estimation for the solution of (49) is less evident and requires a much more detailed analysis. For the sake of simplification, let us also introduce $B(\cdot,\cdot)$ and $F_{(1)}(\cdot)$ as follows:

$$B\left(U_{(1)}^{k+1}, v\right) := \left(\widehat{R}_{(1)}^{k+1}(x)U_{(1)}^{k+1}, v\right) + \alpha_0\left(\frac{\partial}{\partial x}U_{(1)}^{k+1}, \frac{\partial}{\partial x}v\right),$$
$$F_{(1)}^{k+1}(v) := \left(f(x,t_{k+1}), v\right) + \left(\widehat{F}_{(1)}^{k+1}(x), v\right).$$

Then the semi-discretized problem (19) can be rewritten in a compact form: Find $U_{(1)}^{k+1} \in H_0^1(\Omega_x)$, such that

$$B\left(U_{(1)}^{k+1}, v\right) = \widehat{F}_{(1)}^{k+1}(v), \quad \text{for all } v \in H_0^1(\Omega_x). \tag{50}$$

We denote the norm $\|\cdot\|_{1,M}$ which is associated to the bilinear form $B_M(\cdot,\cdot)$:

$$\|\Upsilon_M\|_{1,M} := B_M^{1/2}\left(\Upsilon_M, \Upsilon_M\right), \quad \forall \Upsilon_M \in \mathbb{P}_M(\Omega_x). \tag{51}$$

It can be easily seen, from (47) and (48), that the norm $\|\cdot\|_{1,M}$ is equivalent to the norm $\|\cdot\|_1$ defined in (2). Now we consider the Legendre spectral approximation based on L1-2 method to find $U_{(2)M}^{k+1} \in P_M^0(\Omega_x)$, as follows:

$$B_M\left(U_{(2)M}^{k+1}, v_M\right) = F_{(2)}^{k+1,M}\left(v_M\right), \quad \text{for all } v_M \in \mathbb{P}_M^0(\Omega_x), \tag{52}$$

where the bilinear form $B_M(\cdot,\cdot)$ is defined by

$$B_M\left(U_{(2)M}^{k+1}, v_M\right) := \left(\widehat{R}_{(2)}^{k+1}(x)U_{(2)M}^{k+1}, v_M\right)_M + \alpha_0\left(\frac{\partial}{\partial x}U_{(2)M}^{k+1}, \frac{\partial}{\partial x}v_M\right)_M,$$

and the functional $F_{(2)}^{k+1,M}(\cdot)$ is given by

$$F_{(2)}^{k+1,M}(v_M) := (f(x,t_{k+1}),v_M)_M + \left(\widehat{F}_{(2)}^{k+1}(x),v_M\right)_M,$$

where $\widehat{R}_{(2)}^{k+1}(x)$ and $\widehat{F}_{(2)}^{k+1}(x)$ are defined in (31) and (32), respectively. Taking into account the norm equivalence (48), the well-posedness of the problem (52) is immediate with the help of the Lax-Milgram lemma. However, rigorous error estimation for the solution of (52) is less evident and requires a much more detailed analysis.

$$B\left(U_{(2)}^{k+1},v\right) := \left(\widehat{R}_{(2)}^{k+1}(x)U_{(2)}^{k+1},v\right) + \alpha_0\left(\frac{\partial}{\partial x}U_{(2)}^{k+1},\frac{\partial}{\partial x}v\right),$$
$$F_{(2)}^{k+1}(v) := (f(x,t_k),v) + \left(\widehat{F}_{(2)}^{k+1}(x),v\right).$$

Then the semi-discretized problem (37) can be rewritten in the compact form: Find $U_{(2)}^{k+1} \in H_0^1(\Omega_x)$, such that

$$B\left(U_{(2)}^{k+1},v\right) = F_{(2)}^{k+1}(v), \quad \text{for all } v \in H_0^1(\Omega_x). \tag{53}$$

We denote by $\|\cdot\|_{1,M}$ the norm associated to the bilinear form $B_M(\cdot,\cdot)$:

$$\|\Upsilon_M\|_{1,M} := B_M^{1/2}(\Upsilon_M,\Upsilon_M), \quad \forall \Upsilon_M \in \mathbb{P}_M(\Omega_x).$$

The norm $\|\cdot\|_{1,M}$ is equivalent to the norm $\|\cdot\|_1$ defined in (2). Now we derive the convergence of the present approach by following the ideas in [53, 54].

Theorem 4.1 *Let* $\left\{U_{(q)M}^k\right\}_{k=0}^N$ *be the solution of the problem (49) and (52) for* $q=1,2$, *respectively. With the initial condition* $U_{(q)M}^0$ *taken as* $I_M U_{(q)}^0$. *Let* $\left\{U_{(q)}^k\right\}_{k=0}^N$ *be the solution of the problem (19) and (37) for* $q=1,2$, *respectively. Suppose* $U_{(q)}^k \in H^m(\Omega_x)\cap H_0^1(\Omega_x), m>1,\ q=1,2$. *Then the following estimate follows, for* $0<\alpha<1$,

$$\left\|U_{(q)}^k - U_{(q)M}^k\right\|_{1,M} \le \begin{cases} c_\alpha(\Delta t)^{-\alpha}M^{1-m}\max\limits_{0\leqslant j\leqslant k}\left\|U_{(q)}^j\right\|_m, & q=1, \quad k=1,2,\ldots,N, \\ c_\alpha(\Delta t)^{-2\alpha}M^{1-m}\max\limits_{0\leqslant j\leqslant k}\left\|U_{(q)}^j\right\|_m, & q=2, \quad k=1,2,\ldots,N, \end{cases} \tag{54}$$

where $\|\cdot\|_{1,M}$ *is the norm associated with the bilinear form* $B_M(\cdot,\cdot)$ *defined in (51) and is equivalent to* $\|\cdot\|_1$. c_α *is a positive constant.*

Proof Let $\mathcal{R}_M$ be any function in $\mathbb{P}_M^0(\Omega_x)$. Denote $\rho_M := U_{(q)M}^{k+1} - \mathcal{R}_M$. Then a straightforward calculation shows

$$B_M(\rho_M,\rho_M) = B\left(U_{(q)}^{k+1} - \mathcal{R}_M, \rho_M\right) + B(\mathcal{R}_M,\rho_M) - B_M(\mathcal{R}_M,\rho_M) + F_{(q)}^{k+1,M}(\rho_M) - F_{(q)}^{k+1}(\rho_M).$$

This gives

$$\|\rho_M\|_{1,M}^2 \leqslant \left\|U_{(q)}^{k+1} - \mathcal{R}_M\right\|_1 \|\rho_M\|_1 + |B(\mathcal{R}_M,\rho_M) - B_M(\mathcal{R}_M,\rho_M)| + \left|F_{(q)}^{k+1}(\rho_M) - F_{(q)}^{k+1,M}(\rho_M)\right|.$$

Note that, for all $\mathcal{R}_M \in \mathbb{P}_{M-1}^0(\Omega_x)$, one has $B(\mathcal{R}_M,\rho_M) = B_M(\mathcal{R}_M,\rho_M)$, and hence

$$\|\rho_M\|_{1,M}^2 \leqslant \left\|U_{(q)}^{k+1} - \mathcal{R}_M\right\|_1 \|\rho_M\|_1 + \left|F_{(q)}^{k+1}(\rho_M) - F_{(q)}^{k+1,M}(\rho_M)\right| \quad \forall \mathcal{R}_M \in \mathbb{P}_{M-1}^0(\Omega_x). \tag{55}$$

Now we estimate the last term of the above inequality for the case $q = 1$. Using the definition of $F_{(1)}^{k+1,M}$ and $F_{(1)}^{k+1}$, we have

$$\begin{aligned}\left|F_{(1)}^{k+1}(\rho_M) - F_{(1)}^{k+1,M}(\rho_M)\right| = \Big| &\left[\left(\mathfrak{R}_{(1)}^{k+1}(x)U_{(1)}^k,\rho_M\right) - \left(\mathfrak{R}_{(1)}^{k+1}(x)U_{(1)M}^k,\rho_M\right)_M\right] \\ &+ b_k\left[\left(U_{(1)}^0,\rho_M\right) - \left(U_{(1)M}^0,\rho_M\right)_M\right] \\ &+ \sum_{j=1}^{k-1}(b_j - b_{j+1})\left[\left(U_{(1)}^{k-j},\rho_M\right) - \left(U_{(1)M}^{k-j},\rho_M\right)_M\right] \\ &- \frac{\lambda\alpha_0\Delta t}{2}\sum_{j=0}^{k-1}\Big[\left(\mathbb{K}(x,t_{k+1},t_j)U_{(1)}^j,\rho_M\right) \\ &- \left(\mathbb{K}(x,t_{k+1},t_j)U_{(1)M}^j,\rho_M\right)_M\Big] \\ &- \frac{\lambda\alpha_0\Delta t}{2}\sum_{j=0}^{k-1}\Big[\left(\mathbb{K}(x,t_{k+1},t_{j+1})U_{(1)}^{j+1}),\rho_M\right) \\ &- \left(\mathbb{K}(x,t_{k+1},t_{j+1})U_{(1)M}^{j+1}),\rho_M\right)_M\Big]\Big|,\end{aligned} \tag{56}$$

where $\mathfrak{R}_{(1)}^{k+1}(x) = \left(1 - b_1 - \frac{\lambda\alpha_0(\Delta t)^{1-\beta}}{2(1-\beta)}K(x,t_{k+1})\right)$. For all $g \in H^m(\Omega_x), m > 1$, and for all $g_M \in \mathbb{P}_M(\Omega_x)$, a simple calculation yields the following:

$$\begin{aligned}(g,\rho_M) - (g_M,\rho_M)_M &\leqslant (2\|g - I_M g\|_1 + \|g - I_{M-1}g\|_1)\|\rho_M\|_1 + \|g - g_M\|_{1,M}\|\rho_M\|_{1,M} \\ &\leqslant \left(cM^{-m}\|g\|_m + \|g - g_M\|_{1,M}\right)\|\rho_M\|_{1,M},\end{aligned}$$

for some positive constant c. Applying this result for $g = U_{(1)}^j, g_M = U_{(1)M}^j$ for all $j = 1, 2, \ldots, k$, and using the same arguments as discussed in Theorem 3.2 for a suf-

ficiently large number of mesh points in the temporal direction, the following can be obtained from (56):

$$\begin{aligned}\left|F_{(1)}(\rho_M)-F^M_{(1)}(\rho_M)\right| \leqslant \Bigg\{ & \left[cM^{-m}\left\|U^k_{(1)}\right\|_m+\left\|U^k_{(1)}-U^k_{(1)M}\right\|_{1,M}\right] \\ & +\sum_{j=1}^{k-1}(b_j-b_{j+1})\left[cM^{-m}\left\|U^{k-j}_{(1)}\right\|_m\right. \\ & \left.+\left\|U^{k-j}_{(1)}-U^{k-j}_{(1)M}\right\|_{1,M}\right] \\ & +b_k\left[cM^{-m}\left\|U^0_{(1)}\right\|_m+\left\|U^0_{(1)}-U^0_{(1)M}\right\|_{1,M}\right]\Bigg\}\|\rho_M\|_{1,M}.\end{aligned}$$

Let $\varepsilon^j_M := U^j_{(1)}-U^j_{(1)M}$. Then by using the last inequality of (11), it yields

$$\begin{aligned}\left|F_{(q)}(\rho_M)-F^M_{(q)}(\rho_M)\right| \leqslant & \left[\left\|\varepsilon^k_M\right\|_{1,M}+\sum_{j=1}^{k-1}(b_j-b_{j+1})\left\|\varepsilon^{k-j}_M\right\|_{1,M}\right. \\ & \left.+b_k\left\|\varepsilon^0_M\right\|_{1,M}+cM^{-m}\max_{0\leqslant j\leqslant k}\left\|U^j_{(q)}\right\|_m\right]\|\rho_M\|_{1,M}.\end{aligned} \tag{57}$$

Combining (55) and (57), and using the norm equivalence, we have

$$\begin{aligned}\|\rho_M\|_{1,M} \leqslant & \left\|\varepsilon^k_M\right\|_{1,M}+\sum_{j=1}^{k-1}(b_j-b_{j+1})\left\|\varepsilon^{k-j}_M\right\|_{1,M}+b_k\left\|\varepsilon^0_M\right\|_{1,M}+cM^{-m}\max_{0\leqslant j\leqslant k}\left\|U^j_{(q)}\right\|_m \\ & +c\left\|U^{k+1}_{(q)}-\mathcal{R}_M\right\|_{1,M} \quad \forall\,\mathcal{R}_M\in\mathbb{P}^0_{M-1}(\Omega_x).\end{aligned}$$

Now use the triangular inequality $\left\|\varepsilon^{k+1}_M\right\|_{1,M} \leqslant \|\rho_M\|_{1,M}+\left\|U^{k+1}_{(q)}-\mathcal{R}_M\right\|_{1,M}$ to have

$$\begin{aligned}\left\|\varepsilon^{k+1}_M\right\|_{1,M} \leqslant & \left\|\varepsilon^k_M\right\|_{1,M}+\sum_{j=1}^{k-1}(b_j-b_{j+1})\left\|\varepsilon^{k-j}_M\right\|_{1,M}+b_k\left\|\varepsilon^0_M\right\|_{1,M}+cM^{-m}\max_{0\leqslant j\leqslant k}\left\|U^j_{(q)}\right\|_m \\ & +c\left\|U^{k+1}_{(q)}-\mathcal{R}_M\right\|_{1,M}, \quad \forall\mathcal{R}_M\in\mathbb{P}^0_{M-1}(\Omega_x).\end{aligned}$$

The above estimate especially holds for $\mathcal{R}_M=\pi^1_{M-1}U^{k+1}_{(q)}$, where π^1_{M-1} is the projection operator from $H^1_0(\Omega_x)$ to $P^0_{M-1}(\Omega_x)$. Then, the above inequality yields the following:

$$\begin{aligned}\left\|\varepsilon_M^{k+1}\right\|_{1,M} &\leqslant \left\|\varepsilon_M^{k}\right\|_{1,M} + \sum_{j=1}^{k-1}\left(b_j - b_{j+1}\right)\left\|\varepsilon_M^{k-j}\right\|_{1,M} \\ &\quad + b_k\left\|\varepsilon_M^{0}\right\|_{1,M} + cM^{-m}\max_{0\leqslant j\leqslant k}\left\|U_{(q)}^{j}\right\|_m + cM^{1-m}\left\|U_{(q)}^{k+1}\right\|_m \\ &\leqslant \left\|\varepsilon_M^{k}\right\|_{1,M} + \sum_{j=1}^{k-1}\left(b_j - b_{j+1}\right)\left\|\varepsilon_M^{k-j}\right\|_{1,M} \\ &\quad + b_k\left\|\varepsilon_M^{0}\right\|_{1,M} + cM^{1-m}\max_{0\leqslant j\leqslant k+1}\left\|U_{(q)}^{j}\right\|_m .\end{aligned}$$

Finally, proceeding in a similar way as proved for Theorem 3.2, one can easily have (54). This completes the proof. □

Remark 8 Theorem 4.1 provides critical information on how the error between the weak formulation of the semi-discrete solution and its discrete spectral-based solution behaves as we increase the degree of the polynomial and refine the discretization. This is a fundamental result in numerical analysis, particularly for spectral methods such as the Legendre spectral method. It tells us that the discrete solution in space converges to the exact solution of the weak formulation for a fixed time step when the semi-discrete solution of the weak formulation belongs to $U_{(q)}^k \in H^m(\Omega_x) \cap H_0^1(\Omega_x), m > 1,\ q = 1, 2..$ In this scenario, the spectral method achieves a convergence rate of $\mathcal{O}(M^{-(m-1)})$ precision in the spatial direction on a fixed time step.

We now present the convergence of the fully discrete solution in the following theorem. Notably, the convergence relies on the selection of mesh parameters, specifically the number of discretization points in the temporal direction and the highest order of the Legendre polynomial in the spatial domain. These aspects are explicitly detailed in the theorem below.

Theorem 4.2 *Let u be the exact solution of (1), $\left\{U_{(q)M}^k\right\}_{k=0}^N$ is the solution of the problem (49) ,and (52) for $q = 1, 2$, respectively, with the initial condition $U_{(q)M}^0 = \pi_M^1 U_{(q)}^0$. Then, for all $u \in H^1\left(\overline{\Omega}_t, H^m(\Omega_x) \cap H_0^1(\Omega_x)\right)$ with $\beta \leq \alpha$, $m \geq 2$, we have*

$$\left\|u(t_k) - U_{(q)M}^k\right\|_1 \leq \begin{cases} \mathcal{C}_{u,\alpha}N^{-(2-\alpha)} + c_\alpha(\Delta t)^{-\alpha}M^{1-m}\max\limits_{0\leqslant j\leqslant k}\left\|U_{(q)}^{j}\right\|_m, & q = 1 \quad k = 1, 2, \ldots, N \\ \mathcal{C}_{u,\alpha}N^{-(2-\alpha)} + c_\alpha(\Delta t)^{-2\alpha}M^{1-m}\max\limits_{0\leqslant j\leqslant k}\left\|U_{(q)}^{j}\right\|_m, & q = 2 \quad k = 1, \\ \mathcal{C}_{u,\alpha}N^{-(2-\beta)} + c_\alpha(\Delta t)^{-2\alpha}M^{1-m}\max\limits_{0\leqslant j\leqslant k}\left\|U_{(q)}^{j}\right\|_m, & q = 2 \quad k = 2, 3, \ldots, N. \end{cases}$$

Proof Using the triangle inequality, we have $\left\|u(t_k) - U^k_{(q)M}\right\|_1 \leq \left\|u(t_k) - U^k_{(q)}\right\|_1 + \left\|U^k_{(q)} - U^k_{(q)M}\right\|_1$. Then, the result follows from Theorems 3.2 and 4.1. □

Note 3 For the case where $q = 1$, from the above theorem, it can be confirmed that the proposed fully discrete scheme, which is based on L1/L1-2 Legendre spectral method, converges if we choose the number of collocation points very carefully so that it satisfies $M^{1-m} \leqslant \mathcal{C}\Delta t$, where $\mathcal{C}$ is some positive constant. In this case, the scheme achieves a $\mathcal{O}(N^{-(1-\alpha)})$ temporal rate of accuracy in the space-time domain. In contrast, if one takes $M^{1-m} \leqslant \mathcal{C}(\Delta t)^2$, the proposed scheme converges with a convergence rate of $\mathcal{O}(N^{-\min\{2-\beta,2-2\alpha\}})$ for $q = 2$. In particular, it gives $\mathcal{O}(N^{-2(1-\alpha)})$ convergence rate for $\beta \leq \alpha$. However, these assumptions are purely mathematical. It can be seen from numerical experiments that the convergences of numerical approximations can be guaranteed without these assumptions.

5 Implementation of the Legendre spectral method

For the sake of completeness, the implementation is briefly described here. Considering problem (49), we express the function $U^{k+1}_{(q)M}$ in terms of the Lagrangian interpolants on the Legendre-Gauss-Lobatto points $\xi_j; j = 0, 1, \ldots, M$ with $\xi_0 = -1, \xi_M = 1$, defined as:

$$U^{k+1}_{(q)M}(x) = \sum_{j=0}^{M} U^{k+1}_{(q)j} h_j(x), \quad \text{where} \quad q = 1, 2, \tag{58}$$

where $U^{k+1}_{(q)j} := U^{k+1}_{(q)M}(\xi_j)$, unknowns of the discrete solution. h_j is the Lagrangian polynomial defined in $\overline{\Omega}_x$, i.e.

$$h_j \in \mathbb{P}_M(\bar{\Omega}_x), \quad h_j(\xi_i) = \delta_{ij}, \quad \forall \quad i, j \in \{0, 1, \ldots, M\},$$

with δ_{ij} is the Kronecker-delta symbol. By bringing (58) into (49), and taking into account the homogeneous Dirichlet boundary condition (i.e., $U^{k+1}_{(q)0} = U^{k+1}_{(q)M} = 0$), and choosing each test function v_M to be $h_i(x), i = 1, 2, \ldots, M-1$, we obtain the following for $q = 1$:

$$\left(\widehat{R}^{k+1}_{(1)} \sum_{j=1}^{M-1} U^{k+1}_{(1)j} h_j, h_i\right)_M + \alpha_0 \left(\frac{\mathrm{d}}{\mathrm{d}x} \sum_{j=1}^{M-1} U^{k+1}_{(1)j} h_j, \frac{d}{dx} h_i\right)_M = F^{k+1,M}_{(1)}(h_i), \quad i = 1, 2, \ldots, M-1.$$

Using the definition of the discrete inner product $(\cdot, \cdot)_M$ to the above system, we get

$$\sum_{j=1}^{M-1} U_{(1)j}^{k+1} \sum_{r=0}^{M} \widehat{R}_{(1)}^{k+1}(\xi_r) h_j(\xi_r) h_i(\xi_r) \omega_r + \alpha_0 \sum_{j=1}^{M-1} U_{(1)j}^{k+1} \sum_{r=0}^{M} \frac{\mathrm{d}h_j}{\mathrm{d}x}(\xi_r) \frac{\mathrm{d}h_i}{\mathrm{d}x}(\xi_r) \omega_r = F_{(1)}^{k+1,M}(h_i),$$

where

$$F_{(1)}^{k+1,M}(h_i) := (f(x, t_{k+1}), h_i)_M + \left(\widehat{F}_{(1)}^{k+1}(x), h_i\right)_M.$$

Again, for $q = 2$, we have

$$\left(\widehat{R}_{(2)}^{k+1} \sum_{j=1}^{M-1} U_{(2)j}^{k+1} h_j, h_i\right)_M + \alpha_0 \left(\frac{\mathrm{d}}{\mathrm{d}x} \sum_{j=1}^{M-1} U_{(2)j}^{k+1} h_j, \frac{d}{dx} h_i\right)_M = F_{(2)}^{k+1,M}(h_i), \quad i = 1, 2, \ldots, M-1.$$

Using the definition of the discrete inner product $(\cdot, \cdot)_M$ to the above system gives

$$\sum_{j=1}^{M-1} U_{(2)j}^{k+1} \sum_{r=0}^{M} \widehat{R}_{(2)}^{k+1}(\xi_r) h_j(\xi_r) h_i(\xi_r) \omega_r + \alpha_0 \sum_{j=1}^{M-1} U_{(2)j}^{k+1} \sum_{r=0}^{M} \frac{\mathrm{d}h_j}{\mathrm{d}x}(\xi_r) \frac{\mathrm{d}h_i}{\mathrm{d}x}(\xi_r) \omega_r = F_{(2)}^{k+1,M}(h_i),$$

where

$$F_{(2)}^{k+1,M}(h_i) := (f(x, t_{k+1}), h_i)_M + \left(\widehat{F}_{(2)}^{k+1}(x), h_i\right)_M.$$

Thus, at each time level t_{k+1} for $k = 0, 1, \ldots, N-1$, we arrive at the following matrix statement of problem (49) and (53) for $q = 1, 2$, respectively.

$$\sum_{j=1}^{M-1} W_{ij} U_{(q)j}^{k+1} = F_i^{k+1}, \quad i = 1, 2, \ldots, M-1, \tag{59}$$

where $F_i^{k+1} = F_{(q)}^{k+1,M}(h_i)$, and for all $i, j \in \{1, 2, \ldots, M-1\}$,

$$W_{ij} = A_{ij} + \alpha_0 B_{ij}, \quad B_{ij} = \sum_{r=0}^{M} D_{ri} D_{rj} \omega_r,$$
$$A_{ij} = \widehat{R}_{(q)}^{k+1}(\xi_i) \omega_i \delta_{ij}, \quad D_{ri} = \frac{\mathrm{d}h_i}{\mathrm{d}x}(\xi_r), \quad D_{rj} = \frac{\mathrm{d}h_j}{\mathrm{d}x}(\xi_r).$$

We remember that α_0 is the positive constant; defined as $\alpha_0 := \Gamma(2-\alpha)(\Delta t)^\alpha$. Therefore, at each time step, we obtain a system of linear algebraic equations with different right-hand-side vector $\left\{F_i^{k+1}\right\}_{i=1}^{M-1}$. Since the matrix W is symmetric positive definite, we choose the conjugate gradient method to solve (59).

6 Numerical experiments

In this section, we demonstrate the effectiveness of the proposed approach by implementing it on numerous test examples including smooth and non-smooth kernels. The non-smooth solution is also considered for numerical validation. All plots and data are generated using MATLAB R2023b software on an AppleM3 Pro (memory: 18 GB, macOS: Sonoma 14.5) system. Let us denote $\{u(x_m,t_n)\}_{m=0,n=0}^{M,N}$ and $\{U_m^n\}_{m=0,n=0}^{M,N}$ as the exact and corresponding numerical solutions of the fully discrete scheme. The $\mathcal{L}^\infty$-error (${}^{\mathcal{L}^\infty}\mathscr{E}_M^N$), the weighted discrete $\mathcal{L}^2$-error (${}^{\mathcal{L}^2}\mathscr{E}_M^N$), and the weighted discrete $\mathcal{H}^1$-error are adopted as a measure to evaluate the disparities between the computed and theoretical solutions, and we define these as follows:

$$\begin{cases} {}^{\mathcal{L}^2}\mathscr{E}_M^N := & \left\{\dfrac{1}{(M+1)(N+1)}\displaystyle\sum_{m=0}^{M}\sum_{n=0}^{N}\omega_m\left|u(x_m,t_n)-U_m^n\right|^2\right\}^{1/2}, \\ {}^{\mathcal{L}^\infty}\mathscr{E}_M^N := & \displaystyle\max_{x\in\Omega_x^M}\max_{t\in\Omega_t^N}\left|u(x_m,t_n)-U_m^n\right|, \\ {}^{\mathcal{H}^1}\mathscr{E}_M^N := & \left\{\dfrac{1}{(M+1)(N+1)}\left[\displaystyle\sum_{m=0}^{M}\sum_{n=0}^{N}\omega_m\left|u(x_m,t_n)-U_m^n\right|^2\right.\right. \\ & \left.\left.+\displaystyle\sum_{m=0}^{M}\sum_{n=0}^{N}\omega_m\left|\nabla(u(x_m,t_n)-U_m^n)\right|^2\right]\right\}^{1/2}, \end{cases}$$

where ω_m for $m=0,1,\ldots,M$ are the weights of the Legendre-Gauss-Lobatto quadrature formula. The corresponding temporal rate of convergence (T-RoC) can be calculated as:

$$\begin{aligned} {}^{\mathcal{L}^2}\mathscr{O}_T(M,N) &:= \log_2\left(\frac{{}^{\mathcal{L}^2}\mathscr{E}_M^N}{{}^{\mathcal{L}^2}\mathscr{E}_M^{2N}}\right),\ {}^{\mathcal{L}^\infty}\mathscr{O}_T(M,N) \\ &:= \log_2\left(\frac{{}^{\mathcal{L}^\infty}\mathscr{E}_M^N}{{}^{\mathcal{L}^\infty}\mathscr{E}_M^{2N}}\right),\ {}^{\mathcal{H}^1}\mathscr{O}_T(M,N) := \log_2\left(\frac{{}^{\mathcal{H}^1}\mathscr{E}_M^N}{{}^{\mathcal{H}^1}\mathscr{E}_M^{2N}}\right) \end{aligned}$$

while the spatial rate of convergence (S-RoC) can be estimated as:

$$^{\mathcal{L}^2}\mathscr{O}_S(M,N) := \log_2\left(\frac{^{\mathcal{L}^2}\mathscr{E}_M^N}{^{\mathcal{L}^2}\mathscr{E}_{2M}^N}\right),\ ^{\mathcal{L}^\infty}\mathscr{O}_S(M,N)$$
$$:= \log_2\left(\frac{^{\mathcal{L}^\infty}\mathscr{E}_M^N}{^{\mathcal{L}^\infty}\mathscr{E}_{2M}^N}\right),\ ^{\mathcal{H}^1}\mathscr{O}_S(M,N) := \log_2\left(\frac{^{\mathcal{H}^1}\mathscr{E}_M^N}{^{\mathcal{H}^1}\mathscr{E}_{2M}^N}\right).$$

To start, we begin with an example involving a smooth kernel whose behavior of the analytical solution is sufficiently smooth in the temporal variable but it has different regularity in the spatial direction depending on the choice of the source function. Here, we consider the spatial domain $\Omega_x := (-1,1)$ and the temporal domain $\Omega_t = (0,1]$ for all the test examples.

Example 6.1 Consider the following time-fractional parabolic integro-PDEs with sufficiently smooth kernels.

$$\frac{\partial^\alpha u(x,t)}{\partial t^\alpha} - \frac{\partial^2 u(x,t)}{\partial x^2} = f(x,t) - \int_0^t (t-s)\sin x\, u(x,s)ds,\ \ (x,t) \in \Omega_x \times \Omega_t,$$

with initial conditions:

$$u(x,0) = 0, \qquad x \in \overline{\Omega}_x,$$

and boundary conditions:

$$u(-1,t) = 0, \quad u(1,t) = 0,\ t \in \overline{\Omega}_t,$$

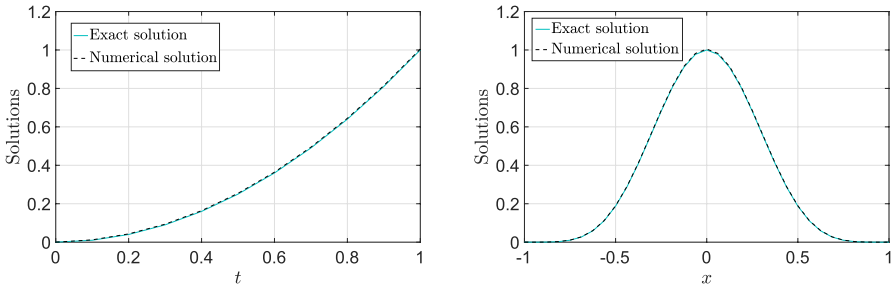
while the exact solution $u(x,t) = t^2(1-x^2)^\nu$ with $\nu \geq 2$, and the source function is given by

$$f(x,t) = \frac{2t^{2-\alpha}}{\Gamma(3-\alpha)}(1-x^2)^\nu - 4\nu(\nu-1)t^2x^2(1-x^2)^{\nu-2} + 2\nu t^2(1-x^2)^{\nu-1} + \frac{t^4}{12}\sin x(1-x^2)^\nu.$$

The investigation of time-fractional parabolic integro-PDEs provides a comprehensive analysis of the performance of both the L1-Legendre and L1-2-Legendre spectral methods. The analytical solution for Example 6.1 exhibits varying degrees of regularity in the spatial variable, depending on the parameter ν. Specifically, the regularity of u can be classified as follows:

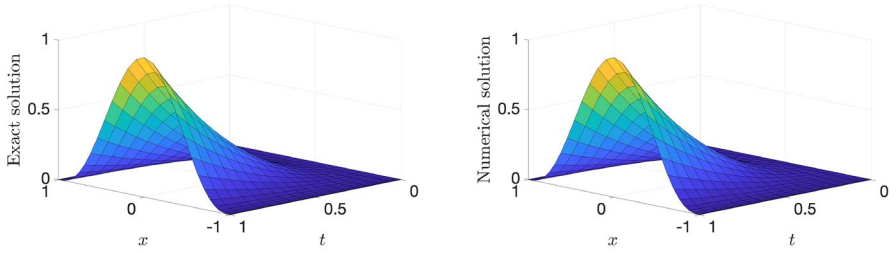
(i) $u \in C^2(\Omega_x)$ iff $2 < \nu < 3$,
(ii) $u \in C^3(\Omega_x)$ iff $3 < \nu < 4$,
(iii) $u \in C^4(\Omega_x)$ iff $4 < \nu < 5$,

and so forth. The results demonstrate a remarkable agreement between the numerical and exact solutions (for example, see the graphs in Fig. 1a & b for the parameters $\alpha = 0.55$ and $\nu = 2.5$), where the solution profiles in the specified time and spatial coordinates show that the numerical solutions closely follow the exact ones, confirm-



(a) $x = 0$. (b) $t = 1$.

Fig. 1 Comparison of solutions with $\alpha = 0.55, \nu = 2.5$ and $N = 10, M = 32$ for Example 6.1 based on L1-Legendre spectral method



(a) Exact solution. (b) Numerical solution.

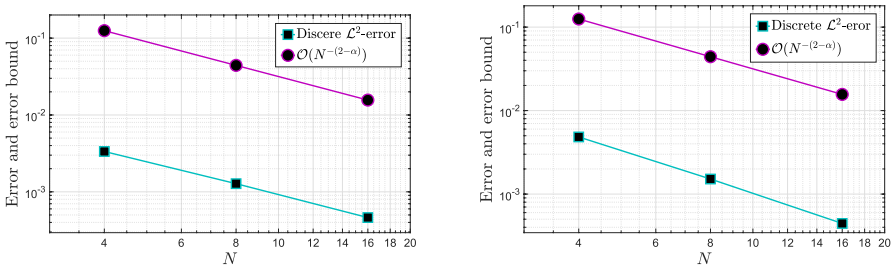
Fig. 2 Surface plots with $\alpha = 0.7, \nu = 2.9$ and $N = 16, M = 32$ for Example 6.1 based on L1-2-Legendre spectral method

Table 2 Error and temporal rate of convergence (T-RoC) with $M = 9, \nu = 2.3$ based on *L1-2-Legendre spectral method* for Example 6.1

α	N	$\mathcal{L}^2$-error	T-RoC	$\mathcal{L}^\infty$-error	T-RoC	$\mathcal{H}^1$-error	T-RoC
0.4	4	3.3273e−3	1.773	2.1915e−2	1.375	4.7303e−2	0.989
	8	9.7371e−4	1.862	8.4491e−3	1.450	2.3838e−2	0.983
	16	2.6784e−4		3.0923e−3		1.2063e−2	
0.6	4	6.8230e−3	1.558	4.1205e−2	1.229	9.1589e−2	0.849
	8	2.3175e−3	1.644	1.7583e−2	1.297	5.0856e−2	0.835
	16	7.4134e−4		7.1553e−3		2.8510e−2	
0.8	4	1.3028e−2	1.275	6.8748e−2	1.070	1.6056e−1	0.687
	8	5.3848e−3	1.336	3.2754e−2	1.130	9.9720e−2	0.673
	16	2.1331e−3		1.4965e−2		6.2554e−2	

ing the precision of the spectral methods. The parabolic nature of the solutions is shown in the surface plots (see Fig. 2).

The convergence analysis reveals a significant finding, which is described in the following. The L1-Legendre spectral method achieves a temporal accuracy rate of $\mathcal{O}(N^{-(2-\alpha)})$, if one has a sufficiently smooth kernel as theoretically discussed in Remark 6. The nonsmooth behavior of the input data can also be considered on the basis of the analysis given in [55]. In this case, one can see from Table 2 that the



(a) L1-Legendre spectral method. (b) L1-2-Legendre spectral method.

Fig. 3 Validation of the theoretical rate of convergence via log-log plot with $\alpha = 0.5, \nu = 2.3, M = 9$ for Example 6.1

Table 1 Error and temporal rate of convergence (T-RoC) with $M = 9$, $\nu = 2.3$ based on *L1-Legendre spectral method* for Example 6.1

α	N	$\mathcal{L}^2$-error	T-RoC	$\mathcal{L}^\infty$-error	T-RoC	$\mathcal{H}^1$-error	T-RoC
0.4	4	2.2245e−3	1.487	8.9474e−3	1.575	1.8335e−2	1.351
	8	7.9370e−4	1.584	3.0034e−3	1.658	7.1851e−3	1.389
	16	2.6466e−4		9.5135e−4		2.7441e−3	
0.6	4	4.8330e−3	1.299	1.9924e−2	1.360	3.9023e−2	1.216
	8	1.9642e−3	1.357	7.7592e−3	1.414	1.6798e−2	1.254
	16	7.6673e−4		2.9108e−3		7.0429e−3	
0.8	4	9.2860e−3	1.114	3.9516e−2	1.159	7.3711e−2	1.069
	8	4.2896e−3	1.156	1.7699e−2	1.191	3.5125e−2	1.115
	16	1.9253e−3		7.7503e−3		1.6211e−2	

Table 3 Error and spatial rate of convergence (S-RoC) with $N = 16$, $\alpha = 0.5$ based on *L1-Legendre spectral method* for Example 6.1

ν	M	$\mathcal{L}^2$-error	S-RoC	$\mathcal{L}^\infty$-error	S-RoC	$\mathcal{H}^1$-error	S-RoC
2.7	2	2.8750e−1	4.844	9.5546e−1	4.520	28.416e−1	4.962
	4	1.0008e−2	4.414	4.1633e−2	4.559	9.1194e−2	4.328
	8	4.6937e−4		1.7656e−3		4.5397e−3	
3.7	2	4.5594e−1	3.695	15.160e−1	2.902	45.110e−1	3.601
	4	3.5212e−2	6.269	2.0287e−1	6.928	3.7169e−1	6.467
	8	4.5663e−4		1.6658e−3		4.2011e−3	
4.7	2	6.2438e−1	2.711	20.765e−1	1.961	61.805e−1	2.612
	4	9.5333e−2	7.752	5.3319e−1	7.757	10.112e−1	7.921
	8	4.4232e−4		2.4642e−3		4.1733e−3	

present numerical simulation through the L1-2 Legendre spectral method leads to the precision $\mathcal{O}(N^{-(2-\alpha)})$. It can be noted that the discretization of the fractional operator reduces the rate of accuracy in time even if the kernel is chosen sufficiently smooth. These theoretical arguments can be confirmed from the log-log plots given in Fig. 3 based on $\mathcal{L}^2$-error. Tables 1, 2, 3, and 4 further substantiate this observation by presenting a detailed error analysis and convergence rates based on the norms $\mathcal{L}^2$, $\mathcal{L}^\infty$ and $\mathcal{H}^1$. Specifically, Tables 1 and 2 demonstrate the temporal rate of con-

Table 4 Error and spatial rate of convergence (S-RoC) with $N = 16$, $\alpha = 0.5$ based on *L1-2-Legendre spectral method* for Example 6.1

ν	M	$\mathcal{L}^2$-error	S-RoC	$\mathcal{L}^\infty$-error	S-RoC	$\mathcal{H}^1$-error	S-RoC
2.7	2	2.8528e−1	4.829	9.5014e−1	4.485	28.302e−1	4.913
	4	1.0038e−2	4.484	4.2414e−2	3.143	9.3919e−2	2.305
	8	4.4860e−4		4.8023e−3		1.9005e−2	
3.7	2	4.5301e−1	3.705	15.089e−1	2.910	44.957e−1	3.605
	4	3.4732e−2	6.407	2.0078e−1	5.503	3.6935e−1	4.420
	8	4.0920e−4		4.4270e−3		1.7258e−2	
4.7	2	6.2073e−1	2.715	20.677e−1	1.963	61.613e−1	2.613
	4	9.4565e−2	7.851	5.3033e−1	7.002	10.069e−1	5.975
	8	4.0943e−4		4.1382e−3		1.6009e−2	

Table 5 A comparative study based on convergence rate and computational cost in seconds with $M = 9, \nu = 4.5$ for Example 6.1

α	N	L1-Legendre spectral method			L1-2-Legendre spectral method		
		$\mathcal{L}^2$-error	T-RoC	CPU time (s)	$\mathcal{L}^2$-error	T-RoC	CPU time (s)
0.3	2	3.1966e−3	1.475	13.850	5.6050e−3	1.663	13.786
	4	1.1500e−3	1.535	26.714	1.7703e−3	1.857	26.942
	8	3.9688e−4		55.808	4.8876e−4		53.984
0.5	2	6.8464e−3	1.316	13.787	1.1034e−2	1.487	14.306
	4	2.7495e−3	1.374	26.832	3.9354e−3	1.673	26.902
	8	1.0611e−3		54.501	1.2341e−3		54.265
0.9	2	2.0063e−2	0.975	13.995	2.8232e−2	0.965	14.510
	4	1.0203e−2	1.019	27.314	1.4459e−2	1.109	27.555
	8	5.0339e−3		57.509	6.7021e−3		54.839

vergence (T-RoC) for different values of α that confirm the accuracy of $\mathcal{O}(N^{-(2-\alpha)})$ in the temporal direction for both schemes in the context of the time-fractional integro-PDEs involving smooth kernels. In contrast, it shows a higher accuracy rate in the spatial direction depending on the regularity of the analytical solution, as shown in Tables 3 and 4. Also, see the theoretical explanation in Theorem 4.1 and further discussion in Remark 8. A comparison is made in Table 5 based on L1 and L1-2 Legendre spectral methods. It can be seen that the L1-2 discretization, while requiring marginally more operations per time step, achieves higher-order accuracy in time, leading to improved overall computational efficiency compared to L1 discretization as discussed in Remark 4.

Now, let us now explore another practical example that features a time-fractional integro-partial differential equation characterized by a weakly singular kernel. This case study will demonstrate the capabilities of our proposed method. When we set $\lambda = 0$, our model reduces to the time-fractional diffusion equation, allowing us to investigate the theoretical convergence and accuracy properties through the lens of the L1-2 discretization method. This special case enables us to establish rigorous convergence results and demonstrate the high accuracy of our numerical approach. In addition, we also take a nonzero λ in this example for numerical demonstrations.

Example 6.2 Suppose $\alpha \in (0,1)$, and consider the time-fractional parabolic integro-PDEs with weakly singular kernels.

$$\frac{\partial^{\alpha} u(x,t)}{\partial t^{\alpha}} - \frac{\partial^2 u(x,t)}{\partial x^2} = f(x,t) - \lambda \int_0^t x(t-s)^{-\beta} u(x,s)ds, \ (x,t) \in \Omega_x \times \Omega_t,$$

with initial conditions:

$$u(x,0) = \phi(x), \ x \in \overline{\Omega}_x,$$

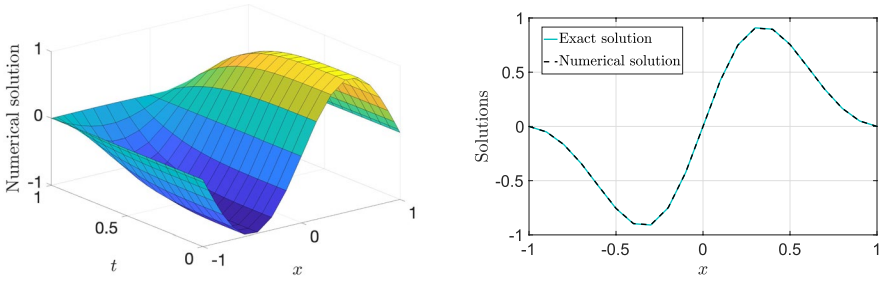
and boundary conditions:

$$u(-1,t) = 0, \quad u(1,t) = 0, \ t \in \overline{\Omega}_t,$$

where $\beta \in (0,1)$. Let us take $\lambda = 1$, $\phi(x) = \sin \pi x$ and the source term $f(x, t)$ is given by

$$f(x,t) = \left[-\frac{\Gamma(4+\alpha)}{\Gamma(4)} t^3 + \pi^2 (1 - t^{3+\alpha}) + \frac{x t^{1-\beta}}{1-\beta} - x t^{4+\alpha-\beta} \boldsymbol{B}(4+\alpha, 1-\beta) \right] \sin \pi x,$$

where $\boldsymbol{B}$ denotes the well-known Euler beta function. The exact solution is given by $u(x,t) = (1 - t^{3+\alpha}) \sin \pi x$. Our investigation now focuses on understanding how the solution dynamics is influenced by varying fractional orders and the presence of the weakly singular kernel. The numerical solution, obtained using the L1-Legendre spectral method, is visualized in Fig. 4a through a surface plot. Furthermore, a cross-sectional view at $t = 0.5$ is presented in Fig. 4b, which highlights the distinctive features of the solution profiles when $\alpha = 0.7$ and $\beta = 0.3$. In the context of the weakly singular time-fractional integro-PDEs, it is important to note that the convergence behavior is highly sensitive to the fractional parameters, as indicated by varying rates across different values of α and β for both spectral methods. For example, the data



(a) Surface plot.

(b) Cross-sectional view at $t = 0.5$.

Fig. 4 Solutions with $\alpha = 0.7, \beta = 0.3$ and $N = 16, M = 20$ for Example 6.2 based on L1-Legendre spectral method

Table 6 Error and temporal rate of convergence (T-RoC) with $M = 9$, $\beta = 0.2$ based on *L1-Legendre spectral method* for Example 6.2

α	N	$\mathcal{L}^2$-error	T-RoC	$\mathcal{L}^\infty$-error	T-RoC	$\mathcal{H}^1$-error	T-RoC
0.3	4	1.0560e−3	1.567	6.9093e−3	1.592	9.2909e−3	1.502
	8	3.5642e−4	1.551	2.2920e−3	1.622	3.2797e−3	1.479
	16	1.2164e−4		7.4440e−4		1.1767e−3	
0.5	4	2.7891e−3	1.407	1.7685e−2	1.374	2.4939e−2	1.341
	8	1.0518e−3	1.429	6.8253e−3	1.421	9.8430e−3	1.393
	16	3.9073e−4		2.5489e−3		3.7479e−3	
0.8	4	8.8480e−3	1.168	5.6221e−2	1.085	8.2104e−2	1.074
	8	3.9382e−3	1.180	2.6502e−2	1.137	3.8986e−2	1.132
	16	1.7387e−3		1.2054e−2		1.7789e−2	

Table 7 Error and temporal rate of convergence (T-RoC) with $M = 9$, $\beta = 0.2$ based on *L1-2-Legendre spectral method* for Example 6.2

α	N	$\mathcal{L}^2$-error	T-RoC	$\mathcal{L}^\infty$-error	T-RoC	$\mathcal{H}^1$-error	T-RoC
0.3	4	3.5250e−4	2.505	2.0597e−3	2.747	3.3269e−3	2.074
	8	6.2081e−5	0.891	3.0688e−4	1.109	7.9029e−4	0.971
	16	3.3468e−5		1.4231e−4		4.0303e−4	
0.5	4	8.8231e−4	2.389	4.9827e−3	2.485	7.4226e−3	2.274
	8	1.6838e−4	1.935	8.9013e−4	2.033	1.5351e−3	1.704
	16	4.4035e−5		2.1745e−4		4.7118e−4	
0.8	4	3.0385e−3	2.061	1.7500e−2	2.090	2.5063e−2	2.032
	8	7.2806e−4	2.105	4.1104e−3	2.184	6.1301e−3	2.064
	16	1.6928e−4		9.0468e−4		1.4661e−3	

presented in Table 6 provides a quantitative assessment of error metrics and T-RoC based on the L1-Legendre spectral method for a fixed $\beta = 0.2$ and varying α that leads to $\mathcal{O}(N^{-(2-\alpha)})$ temporal accuracy provided $\beta \leq \alpha$ as elaborated theoretically in Theorem 3.2 and further discussion in Remark 5. In contrast, the L1-2-Legendre spectral method can lead to $\mathcal{O}(N^{-(2-\beta)})$ temporal accuracy, which will be visible for any $t_n;\ n \geq 2$. This is evident from the results in Table 7. This distinctive behavior is further supported in Tables 8 and 9 for a fixed $\alpha = 0.4$ and for different values of β. It also highlights the convergence rates for $\beta > \alpha$ leading to the temporal accuracy of $\mathcal{O}(N^{-(2-\beta)})$, as depicted at the bottom of these tables. This variation is likely due to the complex interplay between the fractional derivatives and the weakly singular kernel, highlighting the importance of their role in the accuracy rate of the approximate solution. Further, Tables 10 and 11 demonstrate the errors and the spatial rate of convergence (S-RoC) for different choices of α and β based on L1-Legendre and L1-2-Legendre spectral methods, respectively.

If we choose $\lambda = 0$, $\phi = 0$ and $f(x,t) = \left[\dfrac{\Gamma(4+\alpha)}{\Gamma(4)}t^3 + \pi^2 t^{3+\alpha}\right]\sin \pi x$, Example 6.2 becomes a time-fractional diffusion problem with $u(x,t) = t^{3+\alpha}\sin \pi x$ as an analytical solution. In this scenario, one can observe that the present L1-2-Legendre spectral method can lead to a higher rate of accuracy in time, possibly

Table 8 Error and temporal rate of convergence (T-RoC) with $M = 9$, $\alpha = 0.4$ based on *L1-Legendre spectral method* for Example 6.2

β	N	$\mathcal{L}^2$-error	T-RoC	$\mathcal{L}^\infty$-error	T-RoC	$\mathcal{H}^1$-error	T-RoC
0.1	4	1.7823e−3	1.490	1.1745e−2	1.470	1.5836e−2	1.435
	8	6.3448e−4	1.512	4.2384e−3	1.514	5.8559e−3	1.484
	16	2.2246e−4		1.4845e−3		2.0940e−3	
0.3	4	1.7571e−3	1.470	1.0467e−2	1.399	1.5480e−2	1.400
	8	6.3424e−4	1.455	3.9682e−3	1.436	5.8672e−3	1.391
	16	2.3139e−4		1.4668e−3		2.2364e−3	
0.5	4	1.9386e−3	1.271	1.4508e−2	1.253	1.8120e−2	1.266
	8	8.0336e−4	1.098	6.0853e−3	1.358	7.5348e−3	1.136
	16	3.7528e−4		2.3746e−3		3.4279e−3	
0.7	4	4.6449e−3	0.987	3.1419e−2	1.082	4.4790e−2	1.038
	8	2.3427e−3	0.917	1.4836e−2	1.182	2.1810e−2	1.065
	16	1.2409e−3		6.5369e−3		1.0421e−2	
0.9	4	3.4643e−2	0.843	2.2368e−1	0.661	3.4770e−1	0.738
	8	1.9314e−2	0.873	1.4147e−1	0.830	2.0844e−1	0.901
	16	1.0545e−2		7.9591e−2		1.1159e−1	

Table 9 Error and temporal rate of convergence (T-RoC) with $M = 9$, $\alpha = 0.4$ based on *L1-2-Legendre spectral method* for Example 6.2

β	N	$\mathcal{L}^2$-error	T-RoC	$\mathcal{L}^\infty$-error	T-RoC	$\mathcal{H}^1$-error	T-RoC
0.1	4	5.9439e−4	2.491	3.6707e−3	2.355	5.3742e−3	2.320
	8	1.0572e−4	2.401	7.1772e−4	2.389	1.0762e−3	2.001
	16	2.0020e−5		1.3703e−4		2.6890e−4	
0.3	4	5.4075e−4	1.932	2.4679e−3	1.747	4.6512e−3	1.813
	8	1.4170e−4	0.791	7.3520e−4	1.239	1.3233e−3	0.795
	16	8.1877e−5		3.1141e−4		7.6276e−4	
0.5	4	1.0919e−3	0.928	7.1331e−3	1.278	1.0099e−2	1.112
	8	5.7381e−4	0.798	2.9415e−3	1.249	4.6737e−3	0.820
	16	3.3002e−4		1.2372e−3		2.6476e−3	
0.7	4	4.5411e−3	0.926	2.4311e−2	0.850	4.3560e−2	0.971
	8	2.3894e−3	0.903	1.3488e−2	1.024	2.2228e−2	1.049
	16	1.2774e−3		6.6339e−3		1.0739e−2	
0.9	4	3.4890e−2	0.824	2.3187e−1	0.657	3.5235e−1	0.723
	8	1.9712e−2	0.872	1.4701e−1	0.839	2.1341e−1	0.903
	16	1.0768e−2		8.2188e−2		1.1414e−1	

$\mathcal{O}(N^{-(3-\alpha)})$ (see Table 13) compared to the L1-Legendre spectral method (which leads to $\mathcal{O}(N^{-(2-\alpha)})$ temporal accuracy as discussed theoretically in Remark 7, see Table 12). Further, a comparison is made in Table 14 illustrating the computational cost and rate of accuracy in time based on L1 and L1-2 Legendre spectral methods. It is observed that the L1-2 discretization, while requiring marginally more operations per time step, achieves higher-order accuracy $\mathcal{O}((\Delta t)^{3-\alpha})$ in time, leading to improved overall computational efficiency. See the theoretical explanation in Remark 4.

Table 10 Error and spatial rate of convergence (S-RoC) with $N = 32$ based on *L1-Legendre spectral method* for Example 6.2

(α, β)	M	$\mathcal{L}^2$-error	S-RoC	$\mathcal{L}^\infty$-error	S-RoC	$\mathcal{H}^1$-error	S-RoC
(0.3, 0.1)	3	1.3976e−1	8.967	2.7591e−1	8.357	2.6600e+0	8.990
	6	2.7927e−4	3.016	8.4145e−4	1.694	5.2322e−3	3.848
	12	3.4532e−5		2.6009e−4		3.6333e−4	
(0.5, 0.3)	3	1.4179e−1	8.611	2.8509e−1	7.866	2.2732e+0	8.822
	6	3.6259e−4	4.788	1.2217e−3	2.982	5.0232e−3	3.516
	12	1.3122e−5		1.5458e−4		4.3912e−4	
(0.8, 0.5)	3	1.4763e−1	7.137	3.1086e−1	5.803	1.7775e+0	7.514
	6	1.0486e−3	2.550	5.5663e−3	1.474	9.7273e−3	2.644
	12	1.7901e−4		5.5097e−3		1.5565e−3	

Table 11 Error and spatial rate of convergence (S-RoC) with $N = 16$ based on *L1-2-Legendre spectral method* for Example 6.2

(α, β)	M	$\mathcal{L}^2$-error	S-RoC	$\mathcal{L}^\infty$-error	S-RoC	$\mathcal{H}^1$-error	S-RoC
(0.4, 0.2)	3	1.4111e−1	9.038	2.8052e−1	8.255	2.3030e+0	9.061
	6	2.6837e−4	3.088	9.1842e−4	2.483	4.3114e−3	3.569
	12	3.1565e−5		1.6431e−4		3.6334e−4	
(0.6, 0.3)	3	1.4829e−1	8.900	2.9191e−1	7.996	2.3391e+0	9.029
	6	3.1038e−4	1.892	1.1433e−3	1.281	4.4778e−3	2.514
	12	8.3640e−5		4.7060e−4		7.8372e−4	
(0.8, 0.1)	3	1.5951e−1	8.612	3.1499e−1	7.890	2.4241e+0	8.977
	6	4.0768e−4	1.470	1.3283e−3	1.498	4.8115e−3	1.761
	12	1.4713e−4		4.7020e−4		1.4192e−3	

Table 12 Error and temporal rate of convergence (T-RoC) with $M = 9$ based on *L1-Legendre spectral method* for Example 6.2 with $\lambda = 0$

α	N	$\mathcal{L}^2$-error	T-RoC	$\mathcal{L}^\infty$-error	T-RoC	$\mathcal{H}^1$-error	T-RoC
0.2	16	5.6599e−5	1.677	3.3365e−4	1.708	5.0785e−4	1.677
	32	1.7699e−5	1.690	1.0214e−4	1.821	1.5885e−4	1.700
	64	5.4833e−6		2.8916e−5		4.8892e−5	
0.4	16	2.2202e−4	1.529	1.3746e−3	1.526	2.0688e−3	1.520
	32	7.6927e−5	1.546	4.7744e−4	1.564	7.2126e−4	1.546
	64	2.6349e−5		1.6146e−4		2.4694e−4	
0.6	16	6.5985e−4	1.363	4.2689e−3	1.347	6.4212e−3	1.347
	32	2.5659e−4	1.374	1.6779e−3	1.371	2.5239e−3	1.368
	64	9.9001e−5		6.4854e−4		9.7773e−4	

The subsequent numerical experiment demonstrates the robustness and effectiveness of our proposed method when applied to time-fractional weakly singular integro-PDEs characterized by solutions exhibiting unbounded time derivatives at the initial time $t = 0$. To address the singularity at $t = 0$, we employ a non-uniform mesh in time that is dense near the origin. The temporally graded mesh can be defined [47] as:

Table 13 Error and temporal rate of convergence (T-RoC) with $M = 9$ based on *L1-2-Legendre spectral method* for Example 6.2 with $\lambda = 0$

α	N	$\mathcal{L}^2$-error	T-RoC	$\mathcal{L}^\infty$-error	T-RoC	$\mathcal{H}^1$-error	T-RoC
0.2	4	1.7581e−4	2.730	7.6705e−4	2.757	1.7265e−3	2.449
	8	2.6508e−5	2.708	1.1347e−4	2.817	3.1627e−4	2.410
	16	4.0555e−6		1.6097e−5		5.9493e−5	
0.4	4	5.3978e−4	2.492	2.3767e−3	2.470	4.6158e−3	2.415
	8	9.5926e−5	2.510	4.2885e−4	2.574	8.6539e−4	2.444
	16	1.6839e−5		7.2001e−5		1.5898e−4	
0.6	4	1.3412e−3	2.268	6.6298e−3	2.260	1.0921e−2	2.242
	8	2.7848e−4	2.314	1.3840e−3	2.342	2.3085e−3	2.303
	16	5.5997e−5		2.7304e−4		4.6768e−4	

Table 14 A comparative study with $M = 9$ based on convergence rate and computational cost in seconds for Example 6.2 with $\lambda = 0$

α	N	L1-Legendre spectral method			L1-2-Legendre spectral method		
		$\mathcal{L}^2$-error	T-RoC	CPU time (s)	$\mathcal{L}^2$-error	T-RoC	CPU time (s)
0.3	4	1.0402e−3	1.550	27.103	3.2216e−4	2.611	26.856
	8	3.5520e−4	1.582	52.732	5.2730e−5	2.611	54.408
	16	1.1864e−4		105.83	8.6300e−6		110.84
0.5	4	2.7757e−3	1.401	26.778	8.6379e−4	2.378	27.336
	8	1.0510e−3	1.428	53.518	1.6618e−4	2.411	53.494
	16	3.9049e−4		111.90	3.1235e−5		109.19
0.9	4	1.2419e−2	1.088	27.416	4.4181e−3	1.950	27.015
	8	5.8429e−3	1.094	53.617	1.1430e−3	2.020	52.721
	16	2.7372e−3		108.50	2.8185e−4		108.46

$$t_n = T\left(\frac{n}{N}\right)^{\varrho}, \quad \text{for} \quad n = 0, 1, \ldots, N, \tag{60}$$

where $\varrho \geq 1$ denotes the grading parameter, which can be chosen in such a way that the proposed spectral method can achieve an optimal rate of accuracy in time. Note that for $\varrho = 1$, the mesh leads to a uniform mesh in time, as defined in (3).

Example 6.3 Let $\alpha \in (0, 1)$. Consider the time-fractional integro-PDEs having time singularity at $t = 0$:

$$\frac{\partial^\alpha u(x,t)}{\partial t^\alpha} - \frac{\partial^2 u(x,t)}{\partial x^2} = f(x,t) - \int_0^t \exp(x)(t-s)^{-1/2}\, u(x,s)ds, \ (x,t) \in \Omega_x \times \Omega_t,$$

with initial conditions:

$$u(x,0) = 0, \ x \in \overline{\Omega}_x,$$

and boundary conditions:

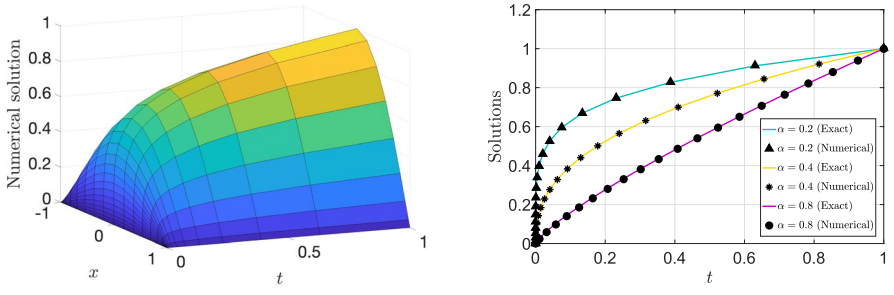
$$u(-1,t)=0,\quad u(1,t)=0,\ t\in\overline{\Omega}_t.$$

Here $f(x, t)$ is given by

$$f(x,t)=\Gamma(1+\alpha)\cos(\pi x/2)+\frac{\pi^2}{4}t^{\alpha}\cos(\pi x/2)+\exp(x)\cos(\pi x/2)t^{\alpha+1/2}\boldsymbol{B}(1+\alpha,1/2).$$

The exact solution to this problem is given by $u(x,t)=t^{\alpha}\cos(\pi x/2)$, which exhibits an unbounded time derivative in a neighborhood of the initial time $t=0$. This leads to the formation of a sharp initial layer near the origin, as clearly depicted in Fig. 5. The sharpness of this singular solution behavior is further emphasized in Fig. 5b, where we observe the evolution of the solution profiles for varying values of the fractional order parameter α. It also highlights that the numerical solution closely aligns with the analytical solution, indicating the excellent accuracy of our proposed technique in capturing the complex dynamics around the singularity. We must note that the location of the layer can vary with respect to time if the input is incompatible with respect to the given model problem [49].

The accuracy of the proposed methodology is evaluated using the $\mathcal{L}^2$ and $\mathcal{L}^{\infty}$ norms. It is important to note that the $\mathcal{H}^1$ norm is not a suitable metric in this case, as the solution exhibits an unbounded time derivative. Let us provide a computational analysis on the performance of the $L1$-Legendre spectral method when applied on a nonuniform temporal mesh for these problems having initial time layers. It is observed that by selecting an appropriate value for the grading parameter, in this case, $\varrho=(2-\alpha)/\alpha$, the method can achieve higher temporal accuracy, as demonstrated in Table 15. In contrast, the use of a uniform mesh ($\varrho=1$) results in larger errors, particularly at lower values of the discretization parameter N, leading to very less accuracy as displayed in Table 16. However, as N increases to 256, a significant reduction in errors is observed, underscoring the benefits of employing a finer mesh configuration to improve the overall accuracy of the solution. While the alternative choices of ϱ are possible, $(2-\alpha)/\alpha$ is the minimum value of ϱ for which the L1 discretization achieves an optimal rate of convergence of order $2-\alpha$. This choice is widely recommended in the literature for problems with weak initial singularities. For



(a) Surface plot. (b) Cross-sectional view at $x=0$.

Fig. 5 Initial layer behaviour with $N=20, M=16$ for Example 6.3 based on L1-Legendre spectral method on non-uniform mesh with $\varrho=(2-\alpha)/\alpha$

Table 15 Error and temporal rate of convergence (T-RoC) for Example 6.3 based on *L1-Legendre spectral* method on *non-uniform mesh* ($\varrho = (2-\alpha)/\alpha$) with $M = 9$

α	N	$\mathcal{L}^2$-error	T-RoC	$\mathcal{L}^\infty$-error	T-RoC
0.3	4	5.3499e−3	1.409	3.1299e−2	1.579
	8	2.0140e−3	1.522	1.0478e−2	1.606
	16	7.0119e−4		3.4409e−3	
0.6	4	3.8685e−3	0.932	2.0226e−2	0.839
	8	2.0274e−3	0.995	1.1306e−2	1.010
	16	1.0175e−3		5.6123e−3	
0.9	4	1.3795e−3	0.522	7.0480e−3	0.262
	8	9.6039e−4	0.408	5.8760e−3	0.604
	16	7.2354e−4		3.8656e−3	

Table 16 Error and temporal rate of convergence (T-RoC) for Example 6.3 based on *L1-Legendre spectral* method on uniform mesh ($\varrho = 1$), with $M = 3$.

α	N	$\mathcal{L}^2$-error	T-RoC	$\mathcal{L}^\infty$-error	T-RoC
0.3	64	3.4379e−3	0.154	2.2657e−2	0.124
	128	3.0906e−3	0.108	2.0791e−2	0.170
	256	2.8675e−3		1.8484e−2	
0.6	64	2.9085e−3	0.190	1.0903e−2	0.510
	128	2.5492e−3	0.103	7.6572e−3	0.442
	256	2.3730e−3		5.6356e−3	
0.9	64	2.2244e−3	0.101	5.9286e−3	0.040
	128	2.0736e−3	0.067	5.7662e−3	0.032
	256	1.9797e−3		5.6398e−3	

Table 17 Error and temporal rate of convergence (T-RoC) for Example 6.3 based on *L1-2-Legendre spectral* method on *non-uniform mesh* ($\varrho = (3-\alpha)/\alpha$), with $M = 9$

α	N	$\mathcal{L}^2$-error	T-RoC	$\mathcal{L}^\infty$-error	T-RoC
0.3	8	3.3589e−3	1.877	2.7550e−2	1.862
	16	9.1473e−4	1.983	7.5779e−3	2.758
	32	2.3131e−4		1.1198e−3	
0.6	8	6.0870e−3	2.013	3.3119e−2	1.965
	16	1.5080e−3	2.442	8.4824e−3	2.228
	32	2.7758e−4		1.8099e−3	
0.9	8	1.5379e−3	1.923	7.7990e−3	2.319
	16	4.0562e−4	1.960	1.5634e−3	1.325
	32	1.0425e−4		6.2404e−4	

instance, see the literature given in [47, 56] and references therein. The convergence can also be guaranteed from Table 18 representing the errors and the spatial rate of convergence (S-RoC) for different values of α based on the L1-Legendre spectral method on non-uniform temporal mesh with $\varrho = 2(2-\alpha)/\alpha$. In the context of the L1-2-Legendre spectral method for solving the time-fractional integro-partial differential model having initial time singularities, the grading parameter $\varrho = (3-\alpha)/\alpha$ is chosen for optimal convergence in time as theoretically discussed in [57]. It can be noticed that the temporal rate of convergence is higher than the rate of convergence obtained by the L1-Legendre spectral method. The computational result is shown in Table 17 for different values of α.

In the following, we consider a two-dimensional weakly singular time-fractional integro-partial differential equation as a test problem to evaluate the accuracy and

Table 18 Error and spatial rate of convergence (S-RoC) for Example 6.3 based on $L1$-Legendre spectral method on non-uniform mesh ($\varrho = 2(2-\alpha)/\alpha$, with $N = 64$)

α	M	$\mathcal{L}^2$-error	S-RoC	$\mathcal{L}^\infty$-error	S-RoC
0.3	2	2.5252e−2	6.841	1.0026e−1	6.502
	4	2.2019e−4	1.105	1.1063e−3	1.006
	8	1.0234e−4		5.5096e−4	
0.6	2	2.6346e−2	6.410	1.0526e−1	6.183
	4	3.0986e−4	1.510	1.4488e−3	1.206
	8	1.0882e−4		6.2803e−4	
0.9	2	2.7409e−2	6.412	1.0538e−1	6.192
	4	3.2195e−4	1.548	1.4413e−3	1.466
	8	1.1007e−4		5.2171e−4	

efficiency of the proposed L1/L1-2 Legendre spectral methods. The problem also exhibits a mild singularity at $t = 0$.

Example 6.4 Consider the time-fractional integro-PDEs in two dimensions having time singularity at $t = 0$:

$$\frac{\partial^\alpha u(x,y,t)}{\partial t^\alpha} - \frac{\partial^2 u(x,y,t)}{\partial x^2} - \frac{\partial^2 u(x,y,t)}{\partial y^2} = f(x,y,t) - \int_0^t (t-s)^{-\beta}\, u(x,y,s)ds,\ (x,y,t) \in \Omega \times \Omega_t,$$

where $\Omega = \Omega_x \times \Omega_y = (-1,1) \times (-1,1)$, with initial condition:

$$u(x,y,0) = 0,\ (x,y) \in \overline{\Omega},$$

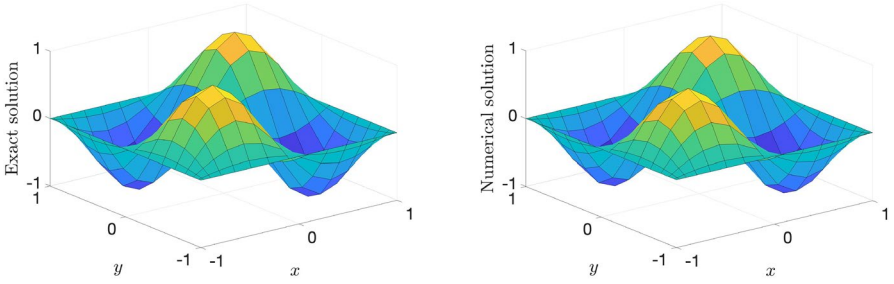
and boundary conditions:

$$\begin{aligned} u(-1,y,t) = 0, &\quad u(1,y,t) = 0,\ (y,t) \in \overline{\Omega}_y \times \overline{\Omega}_t, \\ u(x,-1,t) = 0, &\quad u(x,1,t) = 0,\ (x,t) \in \overline{\Omega}_x \times \overline{\Omega}_t. \end{aligned}$$

Here $f(x, y, t)$ is given by

$$f(x,y,t) = \big(\Gamma(1+\alpha) + 2\pi^2 t^\alpha + t^{1+\alpha-\beta}\boldsymbol{B}(1+\alpha, 1-\beta)\big)\sin(\pi x)\sin(\pi y).$$

The exact solution is given by $u = t^\alpha \sin(\pi x)\sin(\pi y)$. The main adaptations made involve the extension of one-dimensional spectral methods to two-dimensional problems through tensor product constructions. The spatial discretization was enhanced by using a meshgrid to create a 2D grid from 1D Gauss-Lobatto-Legendre (GLL) points, effectively forming a tensor product Legendre spectral scheme where the computational domain is discretized using the Cartesian product of 1D spectral nodes. The basis functions were constructed as 2D tensor products $\widetilde{h}_{i,j}(x,y) = h_i(x) \times h_j(y)$, where h_i and h_j are 1D Legendre polynomials, allowing the 2D solution to be represented as a linear combination of these separable basis functions that maintain the high-order accuracy properties of spectral methods in both spatial dimensions, as shown in Table 21 illustrating the error and the rate of convergence in the spatial direc-



(a) Exact solution. (b) Numerical solution.

Fig. 6 Graphical representation with $\alpha = 0.6, \beta = 0.7$ and $N = 8, M_1 = M_2 = 9$ for Example 6.4 based on L1-Legendre spectral method on non-uniform mesh with $\varrho = 2(2-\alpha)/\alpha$ at $t = 1$

Table 19 Error and temporal rate of convergence (T-RoC) for Example 6.4 based on *L1-Legendre spectral* method with $\beta = 0.3$ and $M_1 = M_2 = 5$

α	N	Graded mesh ($\varrho = (2-\alpha)/\alpha$)				Uniform mesh ($\varrho = 1$)			
		$\mathcal{L}^2$-error	T-RoC	$\mathcal{L}^\infty$-error	T-RoC	$\mathcal{L}^2$-error	T-RoC	$\mathcal{L}^\infty$-error	T-RoC
0.45	4	5.9062e−3	0.737	3.8958e−2	0.126	6.1532e−3	0.534	3.2421e−2	0.089
	8	3.5429e−3	0.692	3.5706e−2	0.124	4.2507e−3	0.537	3.0477e−2	0.059
	16	2.1928e−3		3.2754e−2		2.9293e−3		2.9248e−2	
0.65	4	5.3477e−3	0.679	3.4206e−2	0.122	5.4924e−3	0.547	3.0609e−2	0.086
	8	3.3389e−3	0.637	3.1436e−2	0.099	3.7581e−3	0.546	2.8827e−2	0.059
	16	2.1465e−3		2.9346e−2		2.5734e−3		2.7665e−2	
0.85	4	4.9945e−3	0.635	3.0738e−2	0.110	5.0543e−3	0.579	2.9275e−2	0.090
	8	3.2165e−3	0.598	2.8470e−2	0.079	3.3834e−3	0.560	2.7502e−2	0.061
	16	2.1254e−3		2.6955e−2		2.2942e−3		2.6356e−2	

tion (S-RoC). The numerical integration was adapted by implementing 2D quadrature rules using tensor products of the 1D GLL weights, where the 2D integration weight at each point in the grid (x_i, y_j) becomes the product $w_i \times w_j$ of the corresponding 1D weights, enabling accurate computation of integrals in the 2D domain while preserving the characteristic of exponential convergence rate of the spectral methods. The graphical representation, displayed in Fig. 6, demonstrates the exact and numerical surface plots at $t = 1$ with $\alpha = 0.6, \beta = 0.7$ and $N = 8, M_1 = M_2 = 9$ based on the L1-Legendre spectral method on non-uniform mesh with $\varrho = 2(2-\alpha)/\alpha$. The error and the T-RoC are shown in Tables 19 and 20 based on the L1-Legendre spectral and L1-2-Legendre spectral methods, respectively.

7 Conclusions

This work introduces a robust numerical framework for solving time-fractional integro-parabolic differential equations involving singular kernels, achieving both accuracy and efficiency through an innovative combination of finite-difference and spectral approaches based on the Legendre polynomials. By discretizing the Caputo fractional derivative with L1 and L1-2 schemes in the time direction and approxi-

Table 20 Error and temporal rate of convergence (T-RoC) for Example 6.4 based on *L1-2-Legendre spectral* method on *non-uniform mesh* ($\varrho = (3-\alpha)/\alpha$), with $\beta = 0.3$ and $M_1 = M_2 = 4$

α	N	$\mathcal{L}^2$-error	T-RoC	$\mathcal{L}^\infty$-error	T-RoC
0.4	4	6.2904e−3	0.645	6.0567e−2	0.670
	8	2.6129e−3	1.267	3.8054e−2	1.438
	16	1.6703e−3		1.4049e−2	
0.6	4	4.1192e−3	0.112	4.2625e−2	0.360
	8	3.8102e−3	0.137	3.3203e−2	0.503
	16	3.4659e−3		2.3433e−2	
0.8	4	1.0139e−2	0.220	1.4625e−1	0.707
	8	8.7059e−3	0.409	8.9614e−2	0.864
	16	6.5564e−3		4.9235e−2	

Table 21 Error and spatial rate of convergence (S-RoC) for Example 6.4 based on *L1-Legendre spectral* method with $\beta = 0.3$ and $N = 4$

α	$M_1 = M_2$	Graded mesh ($\varrho = (2-\alpha)/\alpha$)				Uniform mesh ($\varrho = 1$)			
		$\mathcal{L}^2$-error	S-RoC	$\mathcal{L}^\infty$-error	S-RoC	$\mathcal{L}^2$-error	S-RoC	$\mathcal{L}^\infty$-error	S-RoC
0.55	3	6.7186e−2	3.998	3.3624e−1	3.139	8.0675e−2	4.321	3.2365e−1	3.412
	6	4.2058e−3	3.441	3.8167e−2	3.289	4.0357e−3	4.257	3.0414e−2	2.923
	12	3.8733e−4		3.9043e−3		2.1104e−4		4.0103e−3	
0.85	3	6.8786e−2	4.340	3.2251e−1	3.463	7.3010e−2	4.419	3.1880e−1	3.564
	6	3.4688e−3	4.115	2.9247e−2	3.050	3.4118e−3	3.289	2.6950e−2	3.258
	12	2.0014e−4		3.5303e−3		3.4916e−4		2.8176e−3	

mating the integral term using composite trapezoidal and midpoint rules, we have developed semi-discrete schemes that effectively handle the non-local aspects of the problem. A detailed analysis illustrates that the proposed scheme is globally consistent with the temporal accuracy of $\mathcal{O}\left(N^{-\min\{2-\alpha,2-\beta\}}\right)$ based on the norm $\mathcal{H}^1$ on a fixed space mesh. Specifically, the Legendre spectral method was employed for efficient approximation in the spatial direction, which leads to a higher rate of accuracy in space, depending on the regularity (in terms of space) of the analytical solution. Moreover, the final L1/L1-2 Legendre spectral method produces the convergence of the numerical solution on a fixed time mesh as the degree of the Legendre polynomial increases for uniformly bounded solutions in the norm $\mathcal{H}_m$. Each of the theoretical arguments is validated with numerous test examples with non-smooth/smooth kernels. The numerical section also highlights the effectiveness of the spectral method for solving the time-fractional weakly singular integro-PDEs having unbounded time derivatives in the neighborhood of $t = 0$.

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Data availability No data was used for the research described in the article.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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